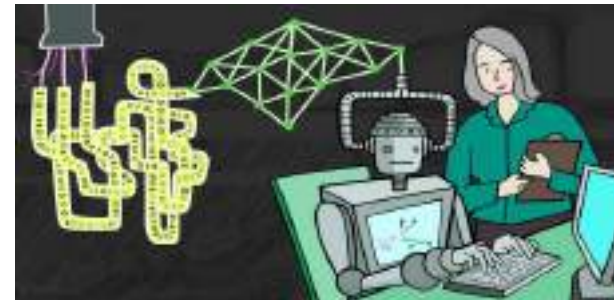


Introduction to MACHINE LEARNING



What is Machine Learning ?

- Machine learning is a branch of artificial intelligence (AI) and computer science which focuses on the use of data and algorithms to imitate the way that humans learn, gradually improving its accuracy.



- It is the use of data to help the computer understand various patterns which helps it make better predictions (better accuracy).

What all will you learn in this course?

Introduction to
Machine
Learning

Applied
Statistics

Basics of Python

EDA

Machine
Learning and its
Types

Projects

Why do we use Machine learning ?

- Machine Learning helps us make better predictions. It is like knowing approximately what's going to happen in the future. Almost every field now used machine learning. Let's Discuss a few.
 - **Stock Market Price Prediction (Finance)**
 - **Recommendation System (Entertainment / Sales)**
 - **Customer Segmentation (All Fields)**
 - **Fraud Detection (Insurance)**

Stock Market Price Predictions (Finance)

- Stock Market has huge variations in the prices every second and tons is generated on a daily basis, understanding the data, deriving useful information and patterns and then making predictions is one of the use cases of Machine Learning.

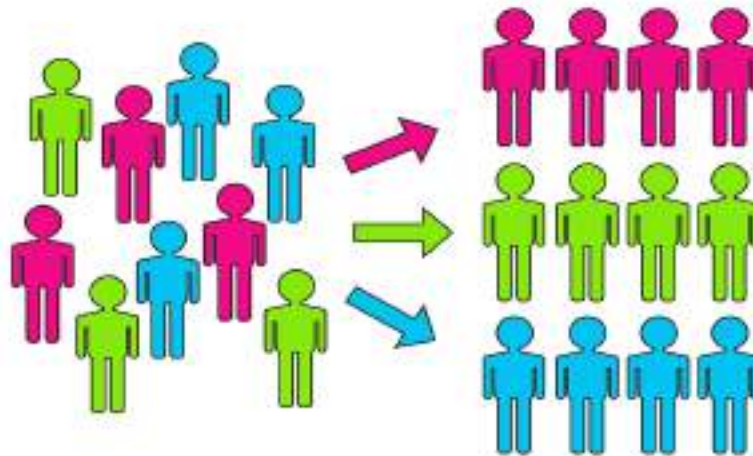


Recommendation System (Entertainment / Sales)

- A system that looks into the various factors it is provided with and then makes similar suggestions based on that.
- The recommendations made by the program, can be judged on the basis of how appealing are they to the customers.
- If it works perfectly, then the GIF on the right comes true.

Customer Segmentation (All Fields)

- This system helps us segregate the customers on the basis of various factors and then cater to them separately so that we can get the best results.
- We again use various factors to identify and determine which customer falls into which group.



Fraud Detection (Insurance)

- The system predicts if a particular transactions seems to be a fraud or not. This is one of the famous use cases for ML.
- With the advancement in technology, this topic has gain a lot of attention. It is becoming normal transactions are difficult to differentiate from the fraudulent ones as the days pass by.



Getting Started

- There are some basic requirements that is necessary before you dive into the Machine Learning aspect of this course.
 - **You must know how to code. (in any language)**
 - **Basic understanding of statistics (applied statistics)**
 - **Basic understanding of mathematics (applied math)**
- If you do not fulfill these requirement at the moment, no need to worry, cause we cover all of these aspects in the course. You will also apply this knowledge in your programs, making your concepts even more clear.

What language is the best to code in?

- For beginners Python has proven to be the most appealing to many. The English like syntax, huge number of libraries to choose from and it being an open source project makes it the ideal choice for beginners.
- Other programming languages can also be used for developing Machine learning applications but for this course we will stick to Python. But I really do appreciate all the people willing to try this out in other programming languages.

Statistics

Descriptive statistics

Probability distributions
[Binomial, Poisson & Normal
distribution]

Probability & Conditional
Probability

Hypothesis Testing

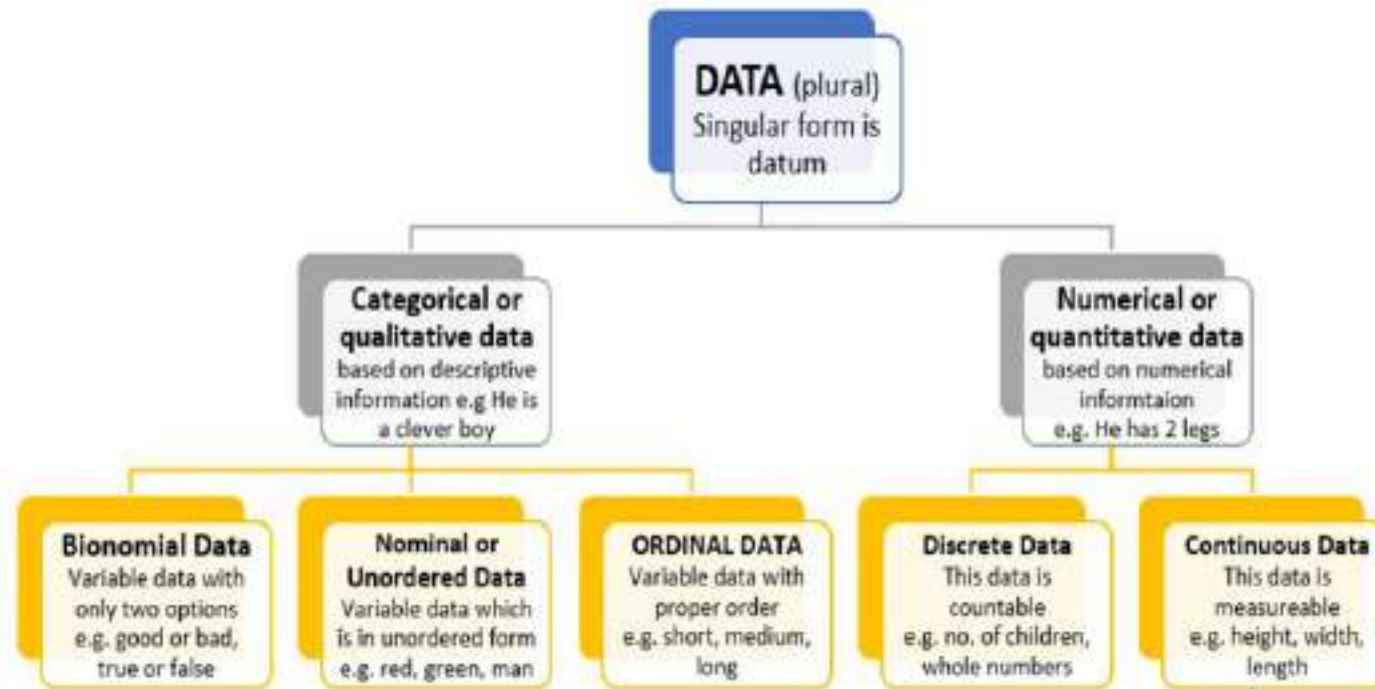
-
- A collection of colorful icons representing various business and technology concepts. The icons include a laptop, a magnifying glass over a lightbulb, a bar chart, a pie chart, a line graph, a folder, a location pin, a printer, a megaphone, and a stack of papers. The icons are arranged in a cluster, with some overlapping, and are set against a background of soft, out-of-focus circles in shades of blue and green.



Why do we need Statistics ?

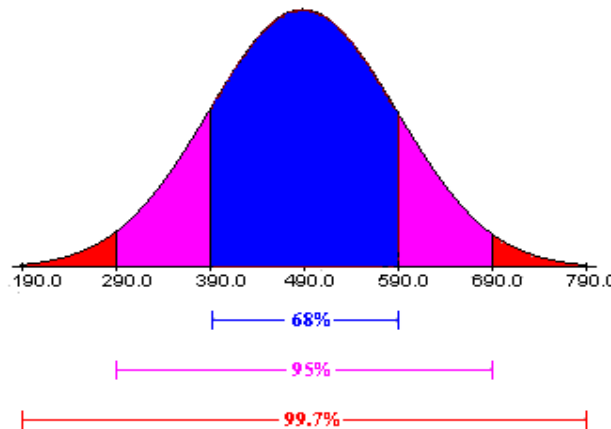
- The major reason is that it helps us understand data better.
- When we understand the data better we work on it better.
- When we work on it better, our deliverables(outcome) after working on the data turns out to be better.

What kind of data do we deal with?



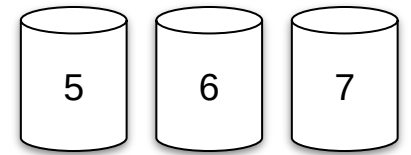
Descriptive Statistics

- Descriptive statistics are brief descriptive coefficients that summarize a given data set, which can be either a representation of the entire or a sample of a population.
- Descriptive statistics are broken down into measures of central tendency and measures of variability (spread)



Measures of central tendency

- When we say measures of central tendency, we basically are talking about 3 things the mean the median and the mode.
- The mean, which is also referred as the average or the expected value. It's nothing but average value of the population or the sample. This value gives us an approximate idea of what the entire data looks like.



$$\text{Mean} = \frac{(5+6+7)}{3} = 6$$

Mean

- Statistical Formula :
- $\bar{X}$ = mean
- $\sum x$ = summation of all the values
- N = number of observations
- Simpler formula :
- Mean = $\frac{\text{Summation of all the values}}{\text{Total number of observations}}$

$$\bar{X} = \frac{\sum X}{N}$$

Examples (Mean)

- There are 3 baskets A, B, C having 12, 14, 22 apples respectively from 3 different locations. The farmer Mr. Orchard wants to know how many apples on an average does he have.
 - Answer = 16
- There are 3 baskets A, B, C having 12, 14, 220 apples respectively from 3 different locations. The farmer Mr. Orchard wants to know how many apples on an average does he have.
 - Answer : 82

Understanding outliers

Person	Income
Mr. X	\$1,500
Mr. Y	\$1,200
Elon Musk	\$1,000,000,000

Mean : \$ 333,334,233.33

Approx : \$ 333,000,000

Median

- Median is the middle observation, after the observations are arranged in the ascending order. In this way it is not affected by the outliers from any direction.

Median

n is odd,

$$\text{Median} = \left(\frac{n+1}{2}\right)^{th} \text{ observation}$$

n is even,

$$\text{Median} = \frac{\left(\frac{n}{2}\right)^{th} + \left(\frac{n}{2} + 1\right)^{th} \text{ observation}}{2}$$

Example (Median)

Person	Income
Mr. X	\$1,500
Mr. Y	\$1,200
Elon Musk	\$1,000,000,000

Mean : \$ 333,334,233.33

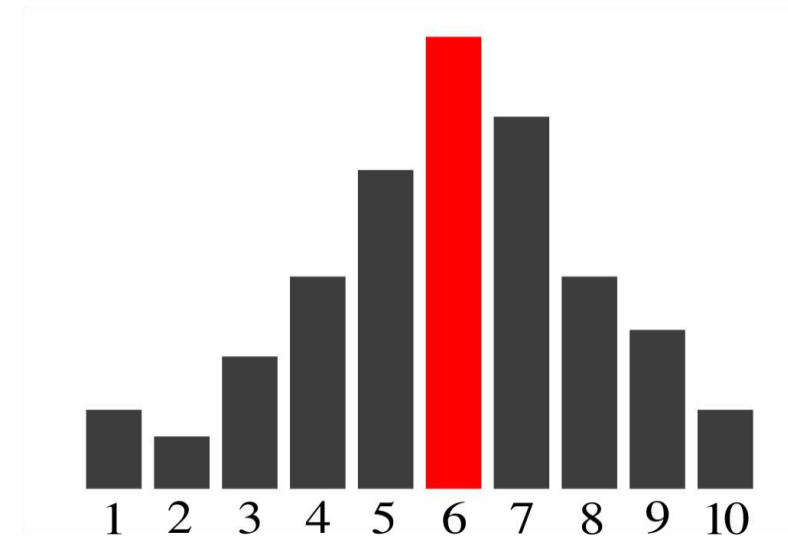
Median : \$ 1,500

Example (Median)

- 4 families meet at a restaurant to have dinner together, the manager asks the host for the average number of people in each family so that he can arrange the tables accordingly. The host recollected:
- Family 1 : 3
- Family 2 : 4
- Family 3 : 5
- Family 4 : 5
- Help the host find the median from the above data.
 - Answer : 4.5

Mode

- Mode is the observation with the highest number of occurrence (frequency) in the data. The data doesn't need to be arranged, we can identify the mode just by glancing over the frequency of each observation.



Example (MODE)

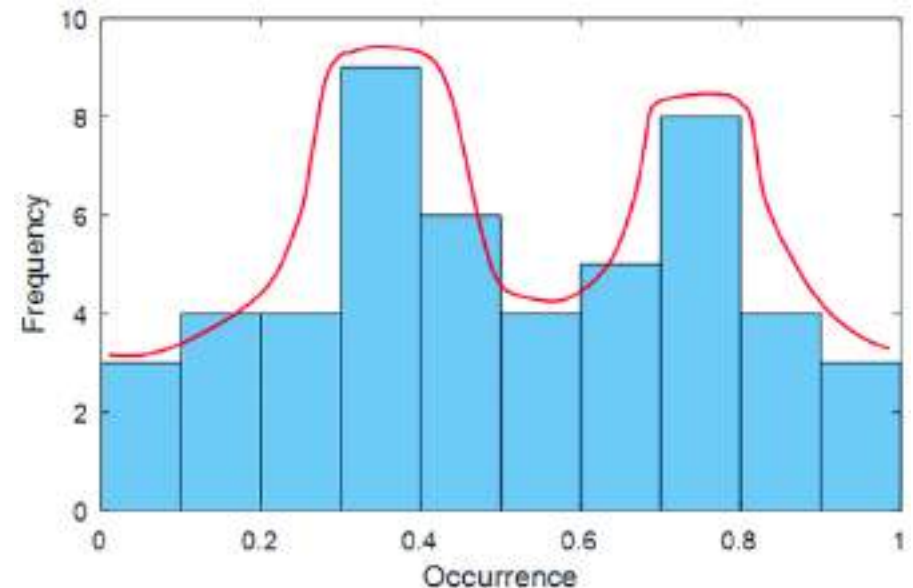
- The amount of money in each person has in his pocket is mentioned below:
- 8, 5, 7, 10, 15, 21, 5, 7, 2, 5
 - 2, 8, 10, 15 and 21 occur 1 time
 - 7 occurs 2 times
 - 5 occurs 3 times
- Answer : The most occurring number is 5, which occurs three times.

Example (MODE)

- 4 families meet at a restaurant to have dinner together, the manager asks the host for the average number of people in each family so that he can arrange the tables accordingly. The host recollected:
- Family 1 : 3
- Family 2 : 4
- Family 3 : 5
- Family 4 : 5
- Help the host find the mode from the above data.
 - Answer : 5

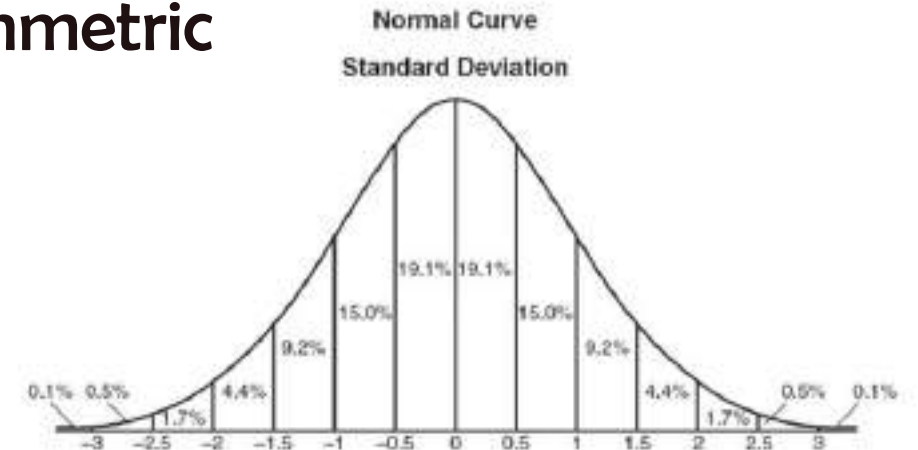
Bimodal

- The data on the right shows that the data has 2 such occurrences that have the high frequencies.
- Example:
The sale of all the beverages from a shop, the data is most likely to look something like this. The sale of coffee and tea are most likely to outrun the other in that section.



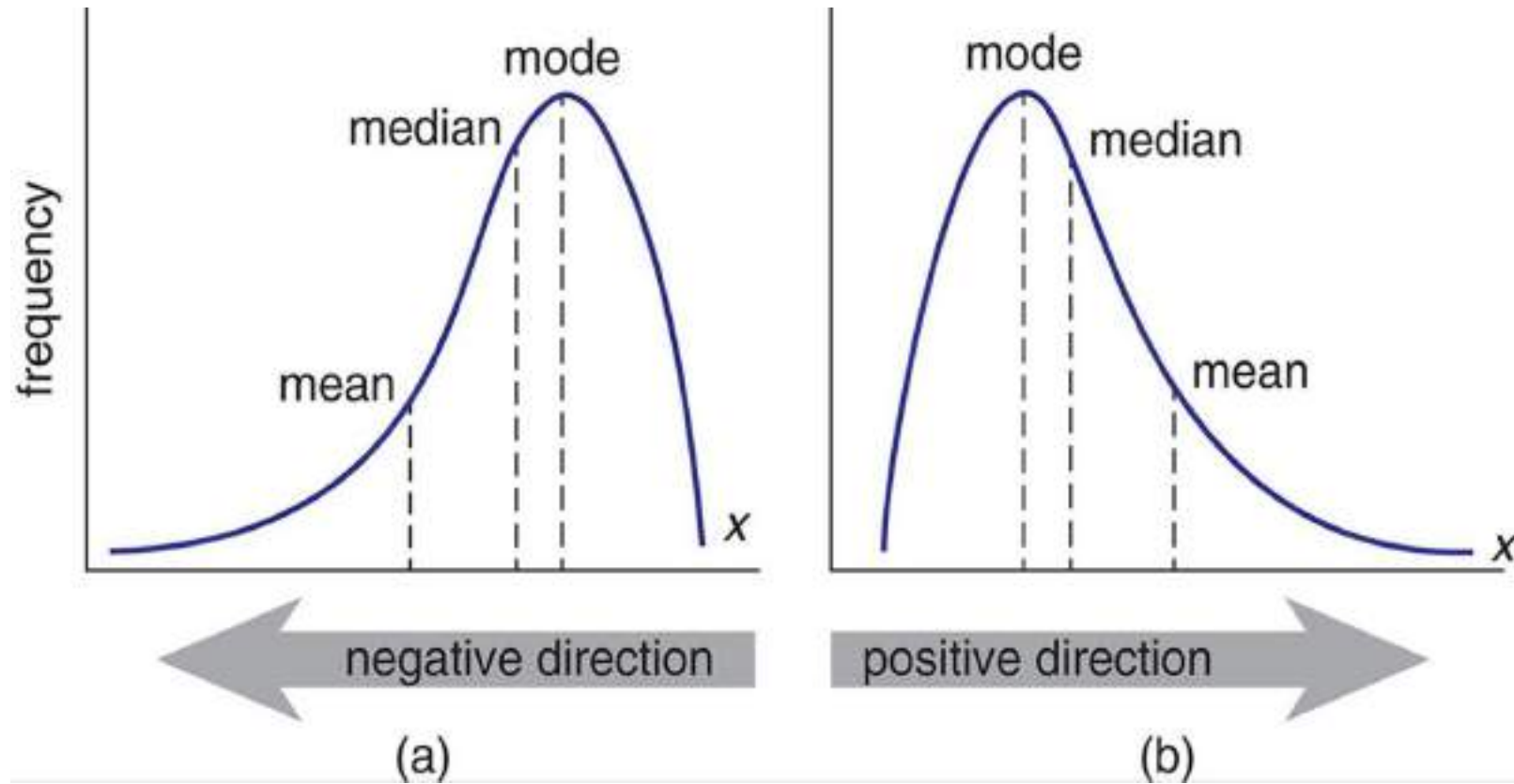
Understanding the bell curve (normal curve)

- The name of the curve is because it's resemblance to a bell.
- The bell shape of the curve portrays the symmetric nature of the curve.



- **Example:**
If we pick up a random group of people from the crowd of all age groups, we will find that most of them will be from the in the age of 30-50, a person having an age of 80+ or a kid of age say 10 is really low.

Skewed Data



**Negatively
Skewed:**

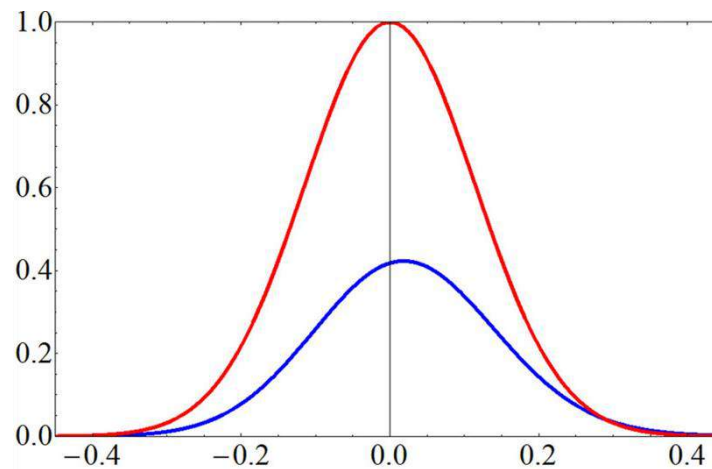
Mode > Median > Mean

**Positively
Skewed:**

Mean > Median > Mode

What is Measure of Spread

- It is a measure that gives us an idea of how spread across the data is.
- Just because 2 data sets have the same mean, doesn't make the data sets similar.



Examples

- There are 2 classes with 5 student's each. The obtained marks are follows:
 - Class 1 : [-10, 0, 10, 20, 30]
 - Class 2 : [08, 09 10, 11, 12]
- The mean for both the classes are 10. Taking that into account can we say that the data sets are similar?
- Answer: No, the data have different spreads.
 - Range for class 1 : $30 - (-10) = 40$
 - Range for class 2 : $12 - 8 = 4$

Why range isn't the best way to understand spreads?

- Like we saw earlier that sometimes the best of methods also fail to give us the correct interpretations of the data set.
- Range is no different, there might be cases where the mean is same the data range is same but the values differ telling us that the data might have a different spread.
- If we go with ranges all the time we might end up with a completely different interpretation.

Variance (σ^2)

Variance Definition in Statistics:

- It is a measure of how data points differ from the mean. According to layman's terms, it is a measure of how far a set of data (numbers) are spread out from their mean (average) value.

Variance Formula

$$\sigma^2 = \frac{\sum (X - \mu)^2}{N}$$

$$s^2 = \frac{\sum (X - \bar{X})^2}{n - 1}$$

Examples

- There are 2 classes with 5 student's each. The obtained marks are follows:
 - Class 1 : [-10, 0, 10, 20, 30]
 - Class 2 : [08, 09 10, 11, 12]
- Variance :
- Class 1 :
$$\frac{(-10-10)^2+(0-10)^2+(10-10)^2+(20-10)^2+(30-10)^2}{5} = \frac{1000}{5} = 200$$
- Class 2 :
$$\frac{(08-10)^2+(09-10)^2+(10-10)^2+(11-10)^2+(12-10)^2}{5} = \frac{10}{5} = 2$$

Standard Deviation (σ)

- The standard deviation is nothing but the root over of variance. It is a quantity expressing by how much the members of a group differ from the mean value for the group.

$$\sigma = \sqrt{\frac{\sum (X - \bar{X})^2}{n - 1}}$$

Examples

- There are 2 classes with 5 student's each. The obtained marks are follows:
 - Class 1 : [-10, 0, 10, 20, 30]
 - Class 2 : [08, 09 10, 11, 12]
- Variance :
- Class 1 :
$$\sqrt{\frac{(-10-10)^2 + (0-10)^2 + (10-10)^2 + (20-10)^2 + (30-10)^2}{5}} = \sqrt{\frac{1000}{5}} = 10\sqrt{2}$$
- Class 2 :
$$\sqrt{\frac{(08-10)^2 + (09-10)^2 + (10-10)^2 + (11-10)^2 + (12-10)^2}{5}} = \sqrt{\frac{10}{5}} = \sqrt{2}$$

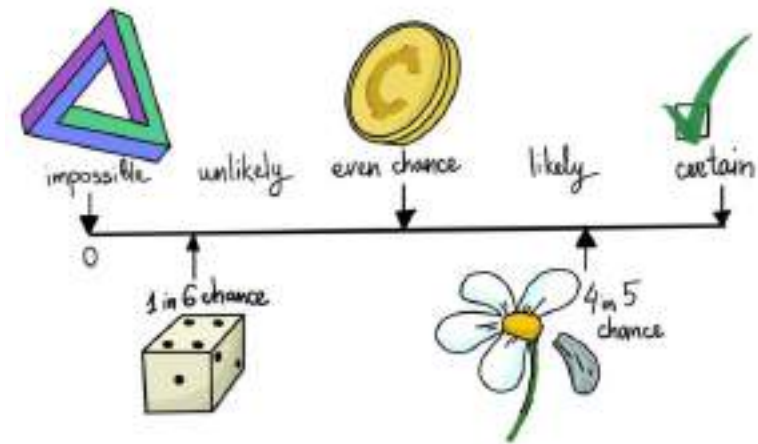
What's the problem with Variance

- Taking the previous example we see that the values for classes are marks. Now when we take the variance into consideration we get the final value in terms of marks². Which doesn't make a lot of sense.

That's the reason we use Standard deviation over variance.

Probability

- The quality or state of being probable; the extent to which something is likely to happen or be the case.
- Basically it tells you the chances of an event to occur or not occur. We can quantify it in terms of a fraction, a decimal or in terms of percentages.
- The value is always between 0 and 1.



So how to calculate it?

- A = event you want to find the probability for.
- $P(A)$ = probability of that event.

$$P(A) = \frac{\text{Number of favorable outcomes to } A}{\text{Total number of outcomes}}$$

Examples

- What is the probability of getting a head for a fair / unbiased coin.
- Ans. A = Getting a head
- $P(A) = \frac{\text{Favorable outcomes}}{\text{Total number of outcomes.}}$
- $P(A) = 1/2$ or 0.5 or 50%

Examples

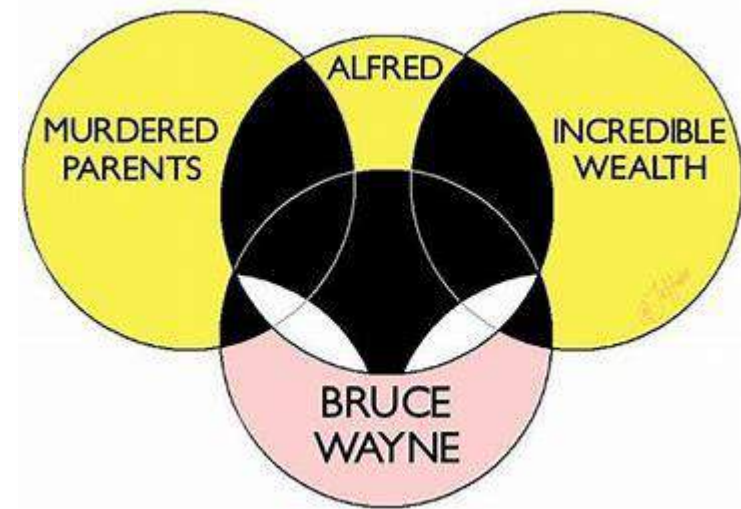
- What is the probability of **not** getting a head for a fair / unbiased coin.
- Ans. A = Not getting a head or getting a tail
- $P(A) = \frac{\text{Favorable outcomes}}{\text{Total number of outcomes.}}$
- $P(A) = 1/2$ or 0.5 or 50%

Examples

- What is the probability of getting a 4 in a roll of a die, considering that it is unbiased.
- Ans. Unbiased die has an equal chance of getting any of its side. Which means that all sides are equally likely to occur.
- A = Getting a 4
- $P(A) = \frac{1}{6}$

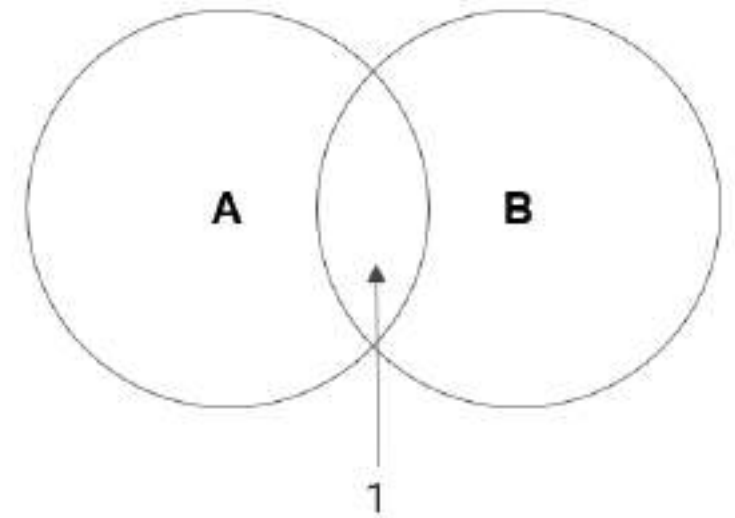
Union and Intersection

- $P(A \cap B)$ = intersection area of event A and B.
 $P(A \cup B)$ = the entire area of A and B.
- $P(A \cap B) = P(A) \cdot P(B)$
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



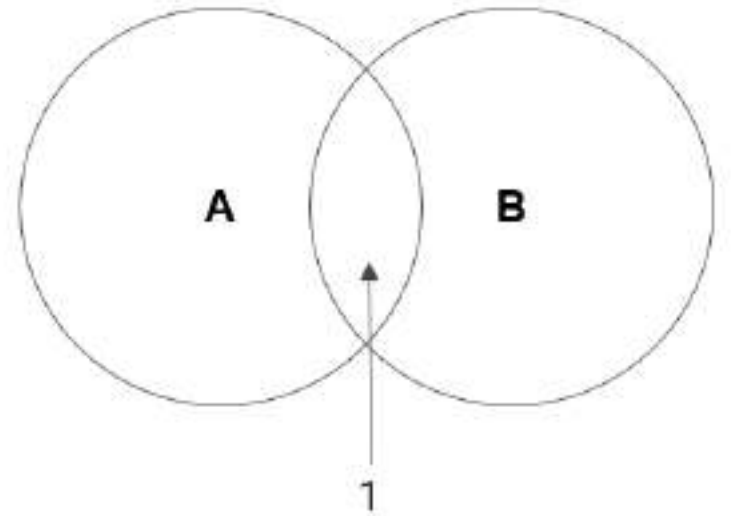
Examples

- There are 52 playing cards in a deck, what is the probability of getting a jack and heart.
- Ans. It's a known fact that there are 4 suits in a deck, and each suit has 1 jack each. So there should be $1/52$. Let's test it out.
- $P(A)$ = probability that a card is jack = $4/52 = 1/13$
 $P(B)$ = probability that a card is heart = $13/52 = 1/4$
- $P(A \cap B) = \frac{1 \times 1}{13 \times 4} = \frac{1}{52}$



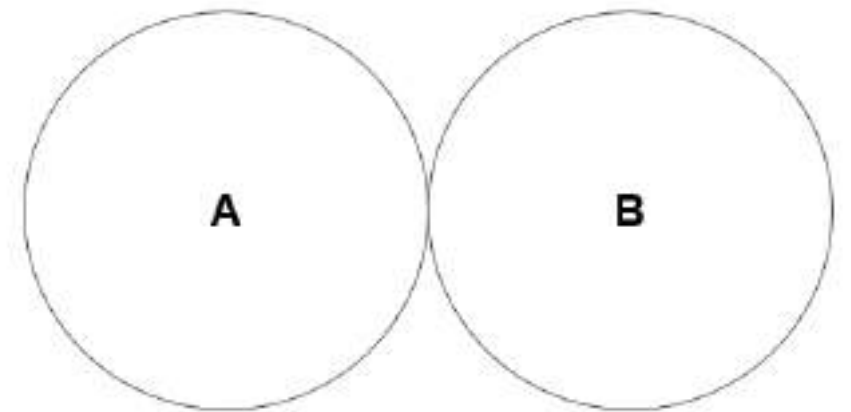
Examples

- There are 52 playing cards in a deck, what is the probability of getting a jack or a heart.
- Ans.
- $P(A)$ = probability that a card is jack = $4/52 = 1/13$
 $P(B)$ = probability that a card is heart = $13/52 = 1/4$
 $P(A \cap B) = 1/52$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $= 1/13 + 1/4 - 1/52$
 $= 16/52 = 4/13$



Examples

- What is the probability of getting a 4 and 6 in a roll of a die, considering that it is unbiased.
- A = event where 4 occurs.
 B = event where 6 occurs.
- $P(A \cap B)$ = Where A and B both occur at the same time.
(which is not possible)
- Since there is no intersection, therefore
- $P(A \cap B) = 0$



Conditional Probability

- Conditional probability is finding the probability of an event given the other event occurs.
- Formula:
$$P(B|A) = P(B \cap A)/P(A)$$
- So basically we try to find the probability of an event based on another event that has already occurred.



Examples

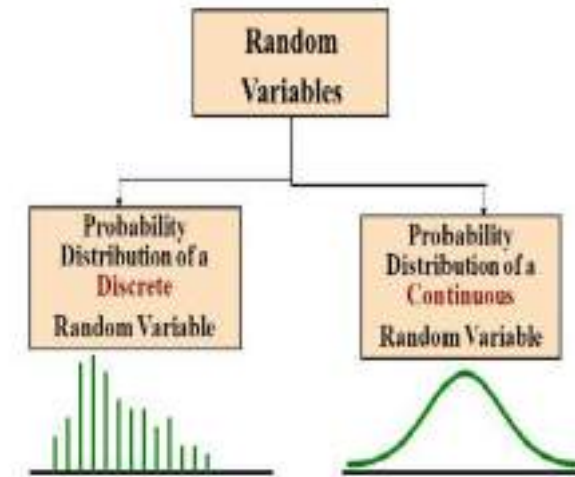
- Two dies are thrown simultaneously, and the sum of the numbers obtained is found to be 7. What is the probability that the number 3 has appeared at least once?
- $A = \{(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (1, 3), (2, 3), (4, 3), (5, 3), (6, 3)\}$
 $B = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$
- $P(A) = 11/36$ $P(B) = 6/36$
- $A \cap B = 2$
- $P(A \cap B) = 2/36$
- Applying the conditional probability formula we get,
 $P(A|B) = P(A \cap B) / P(B) = (2/36) / (6/36) = 1/3$

Examples

- In a group of 100 computer buyers, 40 bought Brand 1, 30 purchased Brand 2, and 20 purchased Brand 1 and 2. If a computer buyer choose at random and bought a Brand 1, what is the probability they also bought a computer from Brand 2?
- $P(A) = 40/100 = 0.4$ $P(A \cap B) = 20/100 = 0.2$
- $P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$ or $\frac{1}{2}$
The Probability that the person bought Brand 2 computer given that he purchased Brand 1 is 50%.

Probability Distributions: Discrete and Continuous

- Random Variables play a vital role in probability distributions and also serve as the base for Probability distributions.
- Probability Distribution is a statistical function which links or lists all the possible outcomes a random variable can take, in any random process, with its corresponding probability of occurrence.



Probability Distributions: Discrete and Continuous

- **Ex: Probability distribution for a discrete random variable.**

No of Heads(X)	Probability P(X)
0	0.25
1	0.5
2	0.25

- If a random variable can take only finite set of values (Discrete Random Variable), then its probability distribution is called as Probability Mass Function or PMF.
- Probability Distribution of discrete random variable is the list of values of different outcomes and their respective probabilities.
- Probability distribution of continuous random variable is called as Probability Density function or PDF.

Need for Probability Distribution

- According to the definition of random variable, it's the variable which can hold different set of values from the outcome of any random process. However, it lacks the capability to capture the probability of getting those different values. So, probability distribution helps to create a clear picture of all the possible set of values with their respective probability of occurrence in any random process.
- Many decisions in business, insurance, and other real-life situations are made by assigning probabilities to all possible outcomes pertaining to the situation and then evaluating the results
 - Saleswoman can compute probability that she will make 0, 1, 2, or 3 or more sales in a single day. Then, she would be able to compute the average number of sales she makes per week, and if she is working on commission, she will be able to approximate her weekly income over a period of time.
 - An investor wants to compare the risks of two different stock options for his portfolio

Binomial Distribution

- A binomial distribution can simply be thought of as the probability of a **SUCCESS** or **FAILURE** outcome in an experiment on survey that is repeated multiple times. The binomial is a type of distribution that has two possible outcomes.
- Ex: A coin toss has only two possible outcomes: heads or tails
Taking a test could have two possible outcomes: pass or fail

Binomial Distribution

Binomial distributions must meet the following three criteria:

- The number of observations or trials is fixed.
- Each observation or trial is independent. In other words, none of the trials have an effect on the probability of the next trial.
- The probability of success (tails, heads, fail or pass) is exactly the same from one trial to another.

$$P(x) = \binom{n}{x} p^x q^{n-x} = \frac{n!}{(n-x)!x!} p^x q^{n-x}$$

where

n = the number of trials (or the number being sampled)

x = the number of successes desired

p = probability of getting a success in one trial

$q = 1 - p$ = the probability of getting a failure in one trial

Poisson Distribution

- Poisson distribution is a probability distribution that is used to show how many times an event is likely to occur over a specified period.
- In other words, it is a count distribution. Poisson distributions are often used to understand independent events that occur at a constant rate within a given interval of time.

Poisson Probability Distribution

$$P(x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

λ - mean number of successes over a given interval

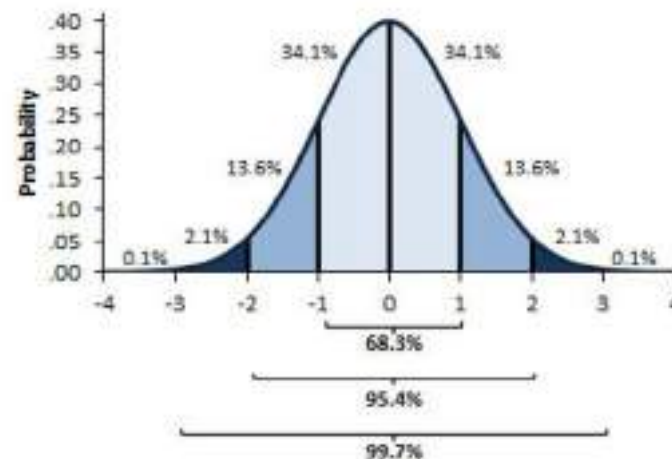
$$\text{Var}(X) = \lambda$$

Poisson Distribution

- It was named after French mathematician Siméon Denis Poisson.
- Ex: If the average number of people who buy cheeseburgers from a fast-food chain on a Friday night at a single restaurant location is 200, a Poisson distribution can answer questions such as, "What is the probability that more than 300 people will buy burgers?"

Normal Distribution

- Normal distribution, also known as the Gaussian distribution, is a probability distribution that is symmetric about the mean, showing that data near the mean are more frequent in occurrence than data far from the mean.
- In graph form, normal distribution will appear as a bell curve.
- The standard normal distribution has two parameters: the mean and the standard deviation.



Normal Distribution

- For a normal distribution, 68% of the observations are within +/- one standard deviation of the mean, 95% are within +/- two standard deviations, and 99.7% are within +/- three standard deviations.
- Many things closely follow a Normal Distribution:
 - Heights of people
 - Size of things produced by machines
 - Errors in measurements
 - Blood pressure
 - Marks on a test

The Normal Distribution:
as mathematical function (pdf)

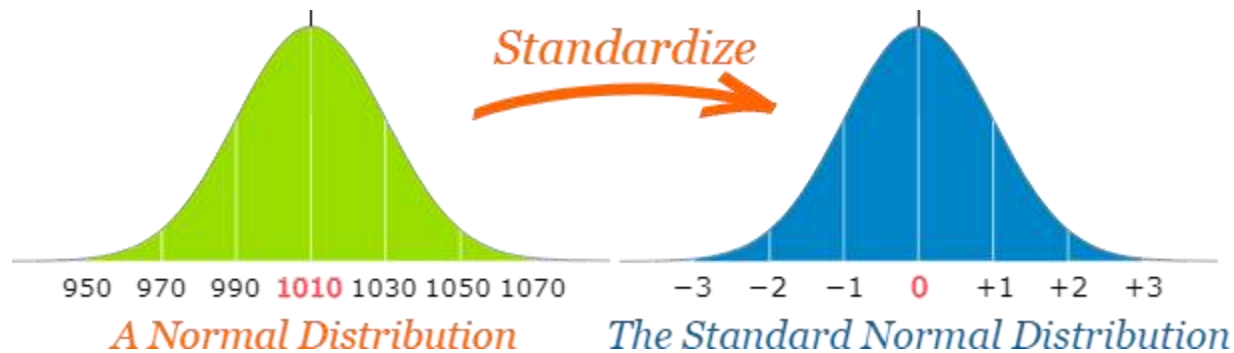
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Note constants:
 $\pi=3.14159$
 $e=2.71828$

This is a bell shaped curve with different centers and spreads depending on μ and σ .

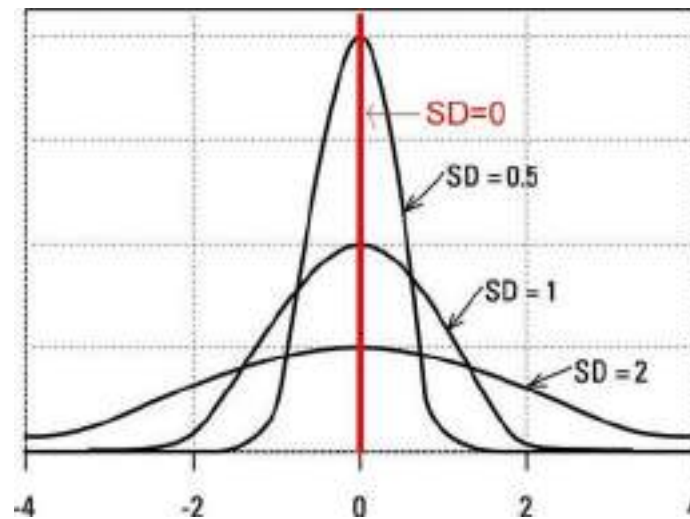
Normal Distribution

- Standard Scores
- The number of standard deviations from the mean is also called the "Standard Score", "sigma" or "z-score"



More about the Normal Curve

- The more congested the data in the middle of the distribution, more data falls within that interval.
- The lesser the data that falls within the interval, the more dispersed the data is.



Hypothesis Testing

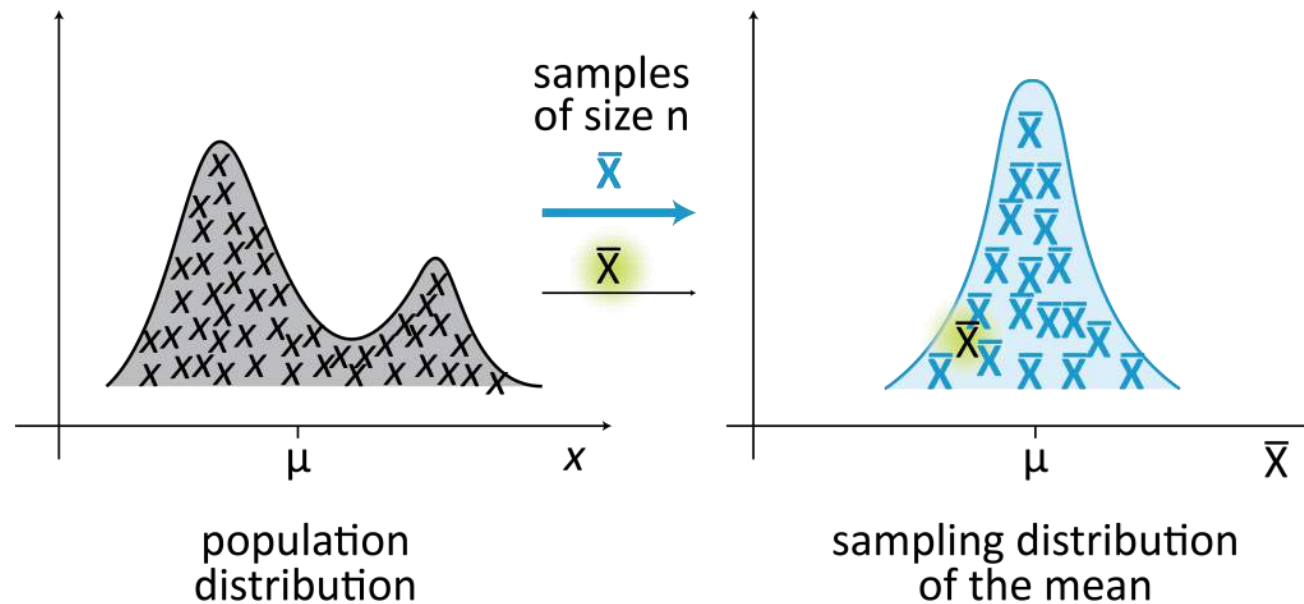
- Hypothesis testing is an act in statistics whereby an analyst tests an assumption regarding a population parameter.
- A Hypothesis Test evaluates two mutually exclusive statements about a population to determine which statement is best supported by the sample data.

Terminologies in Hypothesis Testing

- **Parameter** - It is a summary description of a fixed characteristic or measure of the target population. A Parameter denotes the true value that would be obtained if a census rather than a sample were undertaken.
- **Ex:** Mean (μ), Variance (σ^2), Standard Deviation (σ), Proportion (π)
- **Statistic** - It is a summary description of a characteristic or measure of the sample. The Sample Statistic is used as an estimate of the population parameter.
- **Ex:** Sample Mean ($\bar{x}$), Sample Variance (S^2), Sample Standard Deviation (S), Sample Proportion (p)

Hypothesis Testing

- **Sampling Distribution** - It is a probability distribution of a statistic obtained through a large number of samples drawn from a specific population.



Hypothesis Testing

- **Standard Error (SE)** - It is very similar to the standard deviation. Both are measures of spread.
- The higher the number, the more spread out the data is. The standard error uses statistics (sample data) standard deviation use parameters (population data).
- The standard error tells us how far our sample statistic (like the sample mean) deviates from the actual population mean. The larger our sample size, the smaller the SE.
- In other words, the larger our sample size, the closer our sample mean is to the actual population mean.

Hypothesis Testing

- **Null Hypothesis (H_0)** - A statement in which no difference or effect is expected.
- If the null hypothesis is not rejected, no changes will be made.
- The word “null” in this context means that it’s a commonly accepted fact that researchers nullify.
- It doesn’t mean that the statement is null itself!
- **Alternate Hypothesis (H_1 or H_a)** - A statement in which some difference or effect is expected.
- Accepting the alternative hypothesis will lead to changes in opinions or actions. It is the opposite of the null hypothesis.

Hypothesis Testing

Null vs. Alternative Hypothesis

Null Hypothesis

$$H_0$$

A statement about a population parameter.

We test the likelihood of this statement being true in order to decide whether to accept or reject our alternative hypothesis.

Can include =, ≤, or ≥ sign.

Alternative Hypothesis

$$H_a$$

A statement that directly contradicts the null hypothesis.

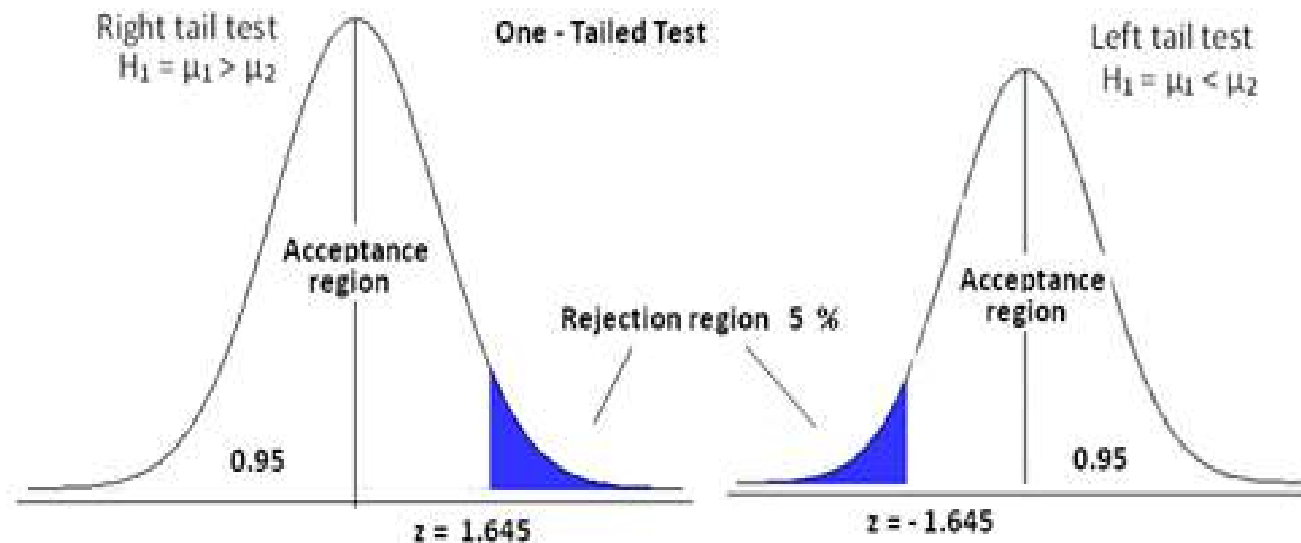
We determine whether or not to accept or reject this statement based on the likelihood of the null (opposite) hypothesis being true.

Can include a ≠, >, or < sign.



Hypothesis Testing

- **One Tailed Test** - It is a statistical hypothesis test in which the critical area of a distribution is one-sided so that it is either greater than or less than a certain value, but not both. If the sample being tested falls into the one-sided critical area, the alternative hypothesis will be accepted instead of the null hypothesis.



Hypothesis Testing

- **Critical Region / Rejection Region** : The critical region is the region of values that corresponds to the rejection of the null hypothesis at some chosen probability level.
- **Two Tailed Test** - It is a method in which the critical area of a distribution is two-sided and tests whether a sample is greater than or less than a certain range of values.
- If the sample being tested falls into either of the critical areas, the alternative hypothesis is accepted instead of the null hypothesis.
- By convention, two-tailed tests are used to determine significance at the 5% level, meaning each side of the distribution is cut at 2.5%.

Hypothesis Testing

- **Test Statistics** - It measures how close the sample has come to the null hypothesis. Its observed value changes randomly from one random sample to a different sample.
- A test statistic contains information about the data that is relevant for deciding whether to reject the null hypothesis or not.
- Different hypothesis tests use different test statistics based on the probability model assumed in the null hypothesis. Common tests and their test statistics include:

Hypothesis test	Test statistic
Z-test	Z-statistic
t-tests	t-statistic
ANOVA	F-statistic
Chi-square tests	Chi-square statistic

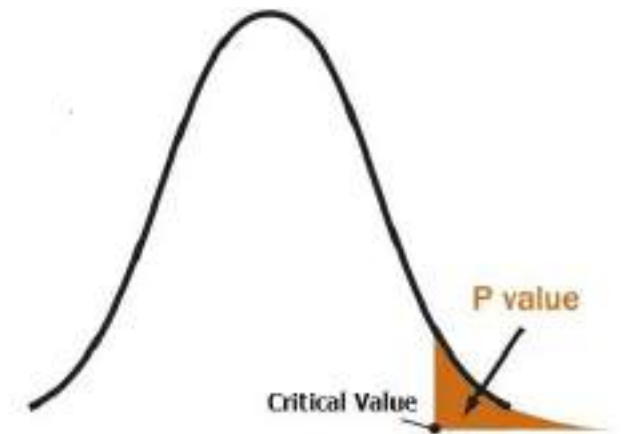
Hypothesis Testing

- **Types of Errors** - There are two types of errors that relate to incorrect conclusions about the null hypothesis.
- **Type I Error** - It occurs when the sample results, lead to the rejection of the null hypothesis when it is in fact true. Type-I errors are equivalent to false positives.
- **Type II Error** - It occurs when based on the sample results, the null hypothesis is not rejected when it is in fact false. Type-II errors are equivalent to false negatives.

		Reality	
		True	False
Measured or Perceived	True	Correct 😊	Type 1 error False Positive
	False	Type 2 error False Negative	Correct 😊

Hypothesis Testing

- **Level of Significance (α)** - It is the probability of making a Type-I error and it is denoted by alpha (α). Alpha is the maximum probability that we have a Type-I error. For a 95% confidence level, the value of alpha is 0.05. This means that there is a 5% probability that we will reject a true null hypothesis.
- **P-Value** - A p-value is used in hypothesis testing to help us support or reject the null hypothesis. The p value is the evidence against a null hypothesis. The smaller the p-value, the stronger the evidence that you should



Diving into Machine Learning



Importing
the data

EDA

Data
Transformation

Model selection

Training the
Model

Testing the
Model

Deployment

Importing the Data



- The data can be in various formats and distributed amongst various files. Combining them is one of the task that is crucial before processing it any further.
- Many a times the data is unstructured, understanding the flow, pattern and converting it into a consistent format is another task that comes under this process.
- Data Cleaning : Sometimes there are missing values in the data that need to be taken care of, there are various of doing so.

Exploratory Data Analysis (EDA)

- Viewing the data in various ways to understand the structure.
- Understanding how the values distributed in the columns
- Use Visual and Non-Visual Methods to understand the data.
- Make a note of all the inferences, insights and assumptions that you gain or make for the visuals.



Data Transformation

- If there is any discrepancy in the data type of the column values or you need to modify the data into a consistent format.
- Sometimes you need to make different column values from the existing ones or remove a few columns as they don't add any information that helps us build a better model.



Model selection

- After understanding how the data is and transforming it in the best way possible for further computation. Choosing the model is an important task. Just because a complex model gives us better accuracy (generally) we don't start of with them.
- We start of with the weak models, and gradually move up the ladder. Each model is selected trained, tested and tuned till we feel that that's the best that the model can do on this data set.



Training the Model

- Each and every model selected is trained over multiple times in multiple ways to understand why one training set is better than the other or why one model performs better over the other.



Testing the Model

- We can only understand how good the model is by testing its learning on data that it has not been trained on.
- The training data has to be unseen by the model to understand how well the model is able to work on data that it hasn't seen before.

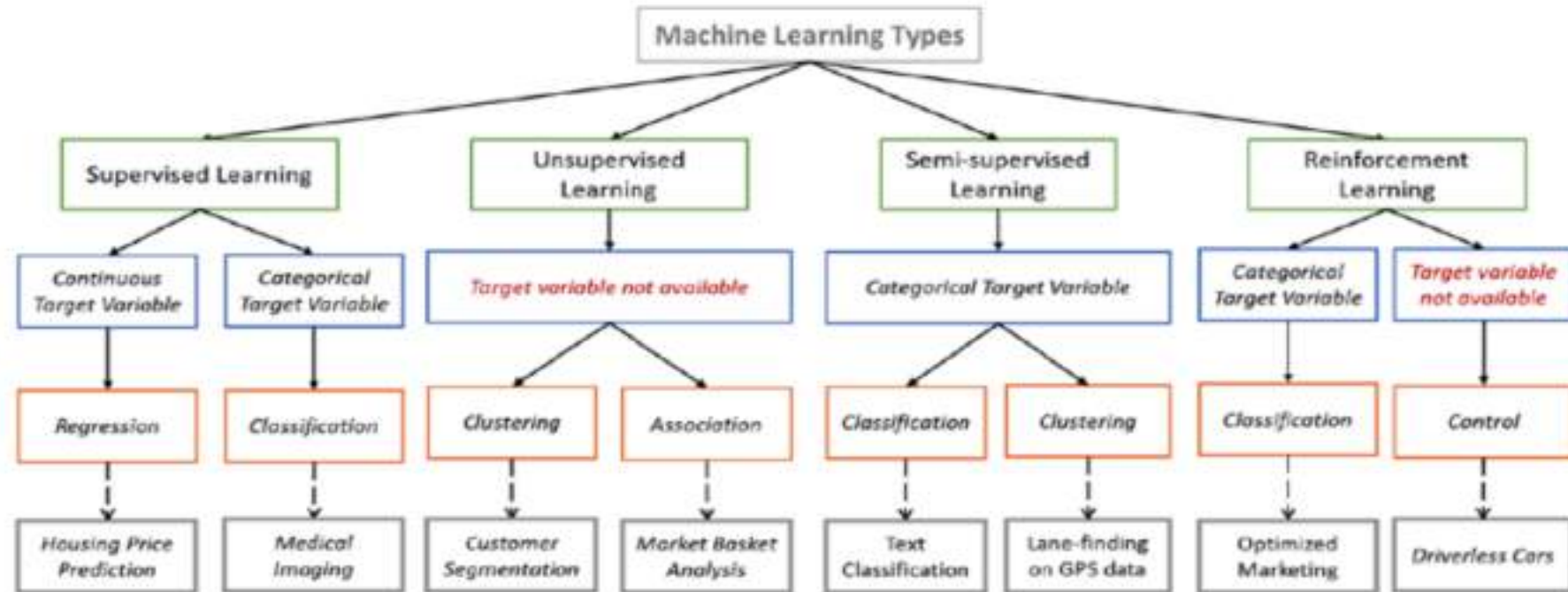


Deployment

- After being satisfied with the results we deploy the model in the real world to see how it works on the data from the real world.
- The task doesn't end here. If the model works fine we try to make it work better. If the model fails to work, then we take it up again and go through the entire cycle all over again. It's a continuous process.



Types of machine learning algorithms



Supervised

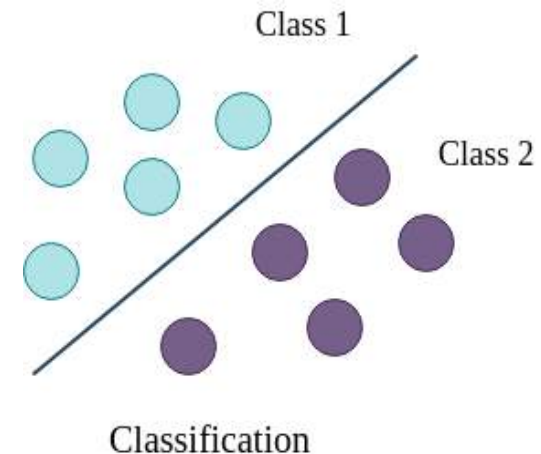
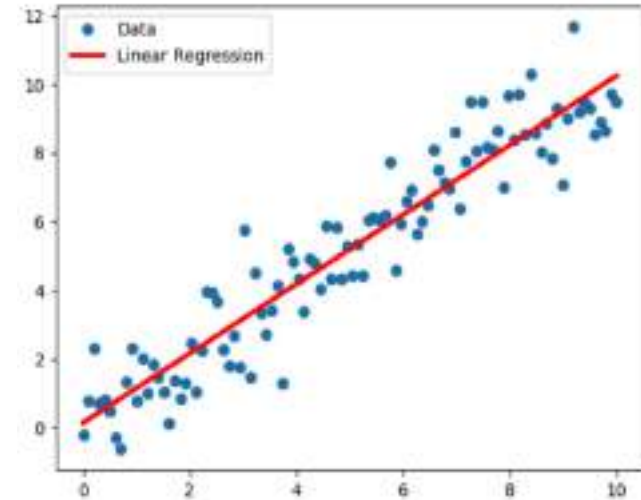
- When the data set that we are working with has labels in it which tells us which all column values represent which category or a continuous value, we perform a supervised task on it.

Regression:

- When the value that you want to predict is of continuous type.

Classification:

- When the value that you trying to predict is a category.



Unsupervised

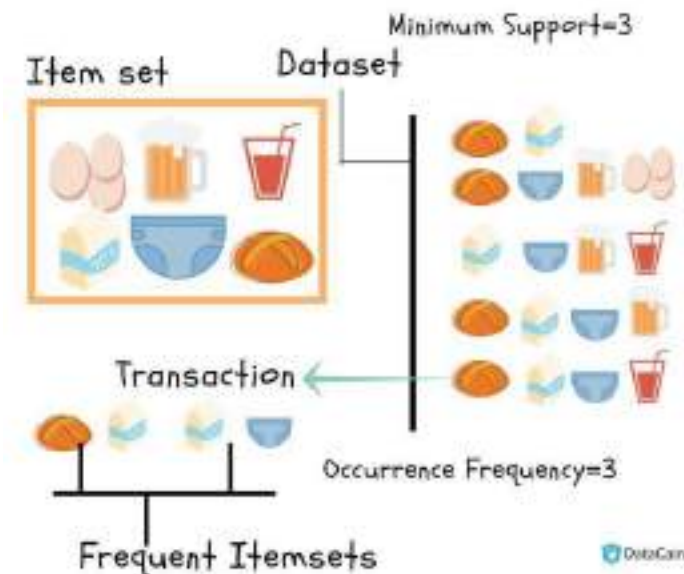
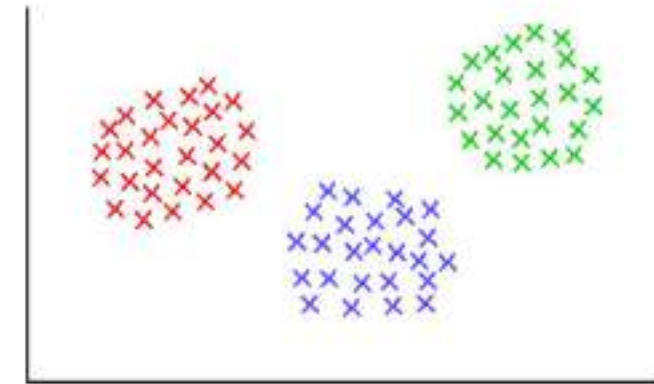
- If the target variable or the value that we are trying to predict is not available, then we perform an unsupervised task.

Clustering :

- Where you group similar points together.

Association:

- Where you try to find a pattern and try to recommend



Semi-supervised

- Getting your hands on labeled data can be really expensive and a tedious task. Getting a dataset with some labels or no labels and then labeling some of them is an easier task.
- Now once you have your data basically you train the model on the labeled dataset and then predict the ones that are not labeled. This reduces the manual labelling time significantly.

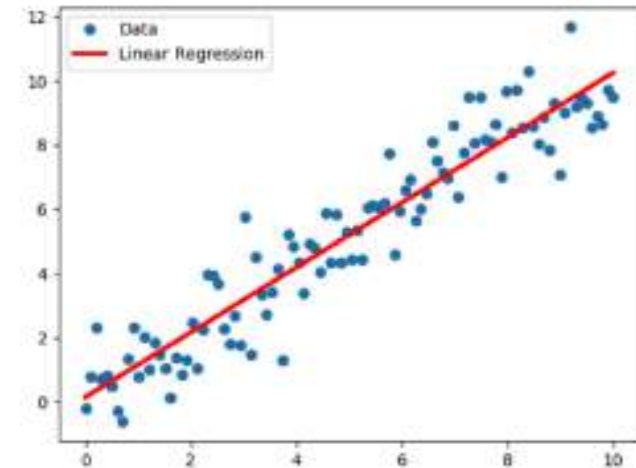
Reinforcement

- Here the learning is a continuous process. Every time the model makes a wrong prediction it is instructed that it made a wrong prediction and that it needs to rectify the same. The cycle goes on till the point the machine can do its assigned task without fail.



Linear Regression

- Linear regression performs the task to predict a dependent variable value (y) based on a given independent variable (x). So, this regression technique finds out a linear relationship between x (input) and y (output). Hence the name, Linear Regression.



Linear Regression

Formula:

- Since we are trying to build a linear relation between the the 2 variables, we use the formula for a straight line.
- The same formula can be written as
 $y = mx + c$.
 - $c = \theta_1$ = Intercept.
 - $m = \theta_2$ = Slope

$$y = \theta_1 + \theta_2 \cdot x$$

How do we update the θ_1 and θ_2 values?

- To find the best fit line we need to have the best θ_1 and θ_2 values. In order to find that we need to minimize the cost function (J). The cost function represents the difference value between predicted and actual.
- Since the predicted and actual difference might have positive and negative values iteratively we square the error to make it positive. (MSE)

$$J = \frac{1}{n} \sum_{i=1}^n (pred_i - y_i)^2$$

How do we understand if the line is best fit.

- θ_1 and θ_2 are randomly selected at first and then optimized using Gradient Descent on the Cost Function. Now when the cost function is minimum we consider that to be the best fit line .

Other ways of evaluation

- **MAE : Mean Absolute Error**, taking the absolute value of the error value.

$$mae = \frac{\sum_{i=1}^n abs(y_i - \lambda(x_i))}{n}$$

- **RMSE: Mean Squared Error** can be difficult to interpret at times when we are dealing with large values. Taking the root of the same gives us better understanding.

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n e_t^2}$$

- In **R2** we first see what is the variation in error terms when we fit a line to the mean of the distribution. Then we fit our line and see how much variation was explained by the new fit line. The higher the variation explained the better is the line. The value is always between 0 and 1.

Other ways of evaluation

- Adjusted R-squared is a modified version of R-squared that has been adjusted for the number of predictors in the model. The adjusted R-squared increases when the new term improves the model more than it would be expected by chance. It decreases when a predictor improves the model by less than expected. Typically, the adjusted R-squared is positive. It is always lower than the R-squared.

$$R^2 = \frac{\text{Var}(\text{mean}) - \text{Var}(\text{line})}{\text{Var}(\text{mean})}$$

$$R^2 = \frac{N \sum xy - \sum x \sum y}{\sqrt{\left[N \sum x^2 - (\sum x)^2 \right] \left[N \sum y^2 - (\sum y)^2 \right]}}$$

Example

Consider the $\text{Var}(\text{mean}) = 32$; and the $\text{Var}(\text{line}) = 6$

- $R^2 = \frac{\text{Var}(\text{mean}) - \text{Var}(\text{line})}{\text{Var}(\text{mean})}$
- $R^2 = \frac{32 - 6}{32}$
- $R^2 = 26/32 = 0.8125$
- That means the line explains 81.25% of the variance, the remaining is considered as error and which can't be explained.

Assumptions to Linear Regression

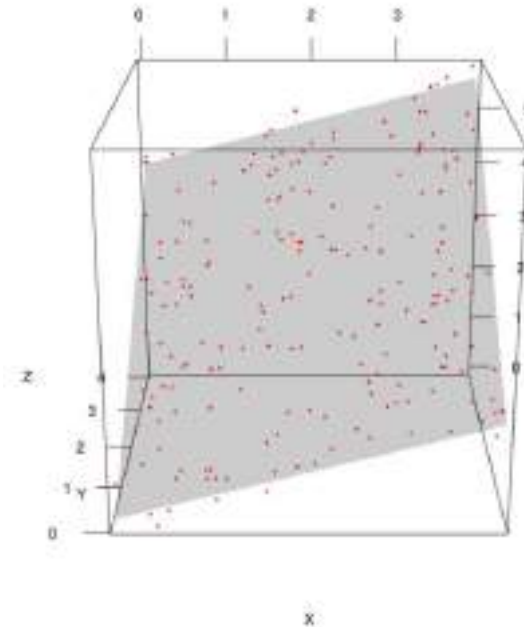
- **Linear relationship:** There exists a linear relationship between the independent variable, x , and the dependent variable, y .
- **Independence:** The residuals are independent. In particular, there is no correlation between consecutive residuals in time series data.
- **Homoscedasticity:** The residuals have constant variance at every level of x .
- **Normality:** The residuals of the model are normally distributed.

If one or more of these assumptions are violated, then the results of our linear regression may be unreliable or even misleading.

Multi Linear Regression

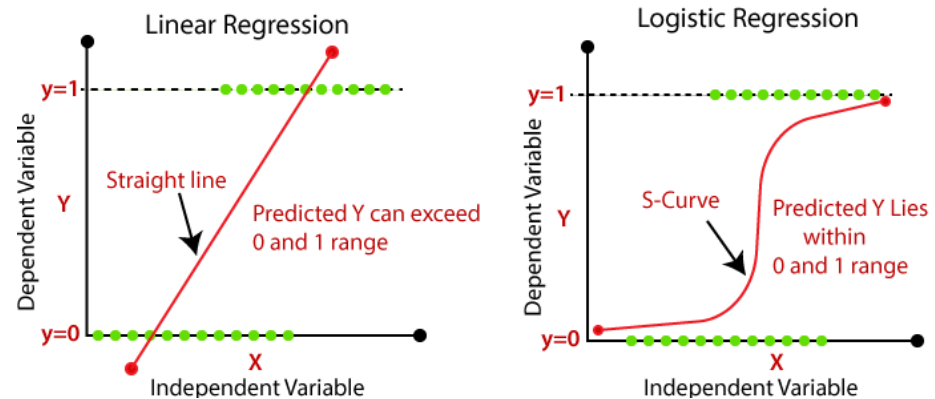
- A line in 2D is a plane in 3D. As the dimensions increase the dimension of the line also increases.
- To take into account all the dependent and the independent variables the line needs to become a plane, in the 3D.

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n + \epsilon$$



Logistic Regression

- Logistic regression is basically a supervised classification algorithm. In this analytics approach, the dependent variable is finite or categorical: either A or B (binary regression) or a range of finite options A, B, C or D (multinomial regression).
- It is used in statistical software to understand the relationship between the dependent variable and one or more independent variables by estimating probabilities using a logistic regression equation.



Evaluate Logistic Regressions

- Confusion matrix is a good way to have a look at the correctly identified classes and misclassified classes.
- Using the values from there we can find the accuracy. The formula for that is total number of correctly classified records divided by the total number of records.

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

Logistic Regression

- **Stratified Sampling** : when there is a class imbalance it's best to use stratified sampling, this makes sure that the test data and train data have an equal distributions in terms of class proportions.

Example:

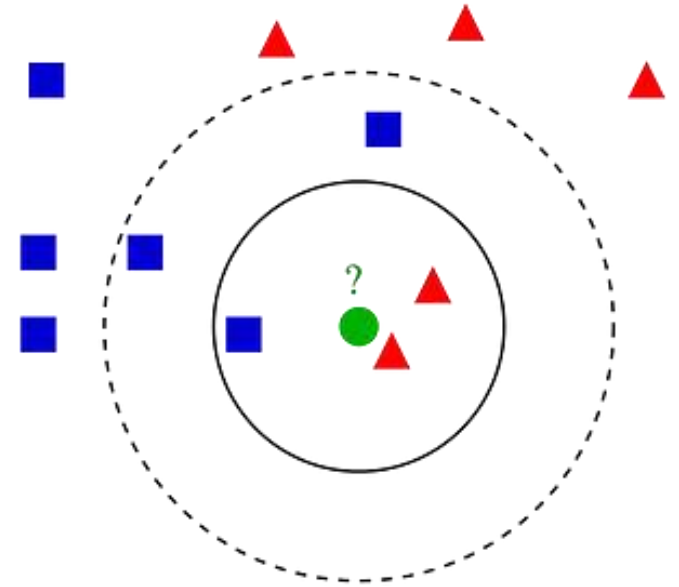
- **Total number of classes** : {0: 100 , 1:50}
- **Considering the test size to be 20%:**
 - Test records for the model {0 : 20, 1:10}

Logistic Regression

- **Types of Logistic Regression:**
 - **Binomial:** In binomial Logistic regression, there can be only two possible types of the dependent variables, such as 0 or 1, Pass or Fail, etc.
 - **Multinomial:** In multinomial Logistic regression, there can be 3 or more possible unordered types of the dependent variable, such as "cat", "dogs", or "sheep"
 - **Ordinal:** In ordinal Logistic regression, there can be 3 or more possible ordered types of dependent variables, such as "low", "Medium", or "High".
- **Assumptions for Logistic Regression:**
 - The dependent variable must be categorical in nature.
 - The independent variable should not have multi-collinearity.

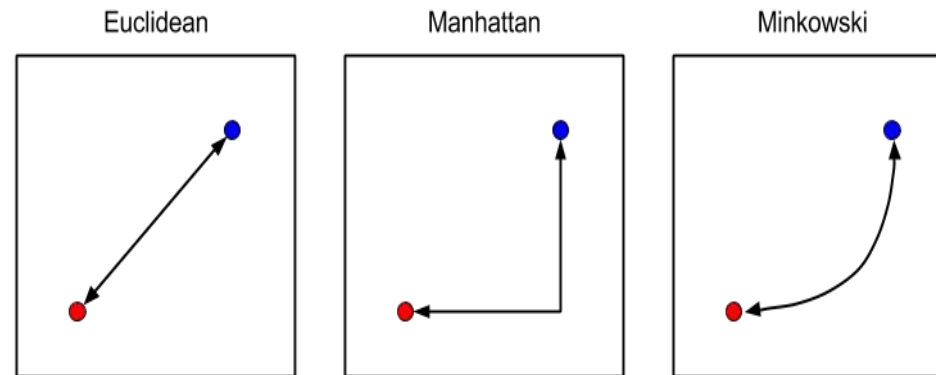
K nearest Neighbours

- KNN stands for K nearest Neighbors. Now k is nothing but a placeholder, which depicts the number of neighbors you want to take into consideration. Example $k = 3$, I am going to take the 3 most nearest neighbors.
- How do we measure which element is close, we use some distance measure to decide that.



Distance Measures for KNN

- **Euclidean** : The distance is calculated through a straight line between two points.
- **Manhattan** : The distance is the summation of the perpendicular distance and horizontal distance.
- **Minkowski** : it's the distance between 2 points by using a curved line.



KNN

- **Model Summary:**
- **Precision :** What proportion of positive identifications was actually correct?
- **Recall :** What proportion of actual positives was identified correctly?
- **F1 Score :** It is calculated from the precision and recall of the test, The F1 score is the harmonic mean of the precision and recall.

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

KNN

- Confusion Matrix:
[TP: 1 , FP: 1
FN: 8, TN: 90]
- Precision = $1/1+1 = 0.5$
- Recall = $1/1+8 = 0.11$
- F1-Score = $2 \times (0.055/0.61) = 0.18$
- Accuracy = $91 / 100 = 0.91$

KNN

- **Grid Search Cross Validation:** This is a hyperparameter tuning method where you put in all the parameter values that you want to train and test your model with and on the basis of that you get a combination of all the values passed. You can select the best out of that.
- **Random Search Cross Validation:** This is similar to Grid Search but doesn't make a combination of all the values. It make a combination of that values that are most likely to give you better results. (Best for larger datasets and more number of parameters)

KNN

- **K Fold and Stratified K Fold Cross Validation:** In K fold we basically fold our data set k times and do the train test on that data set. This gives us K number of accuracies, now you can either look at the range in which the accuracies lie or just have an average accuracy.
- **Stratified K Fold CV** is just an extension of the same old K fold, the only difference is that the data over here is stratified, has an equal proportion of class distribution in both the training and the testing splits.

Naive Bayes

- Naive Bayes uses the Bayes' Theorem and assumes that all predictors are independent. In other words, this classifier assumes that the presence of one particular feature in a class doesn't affect the presence of another one.
- The very reason the name of the classification technique is naive is because of the same reason that the model assumes that the features are conditionally independent.

Bayes Theorem

- If A and B are two events, then the formula for Bayes theorem is given by:
- Conditional Probability:
 - $P(A|B) = P(A \cap B) / P(B)$
- Bayes' Formula:
 - $P(B|A) = (P(A|B) \times P(B)) / P(A)$
 - $P(B|A) = ((P(A \cap B) / P(B)) \times P(B)) / P(A)$
 - $P(B|A) = P(A \cap B) / P(A)$

Applying Bayes Theorem to Understand Conditional Independence

Q. Let event A signify the occurrence of head and event B be the occurrence of a 1 in a roll of a dice. Find the Probability of B given that A occurs.

- $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{6}$
 $P(A \cap B) = P(A) \times P(B)$ [As there is no intersection points]
- $P(B|A) = P(A \cap B) / P(A)$
 $P(B|A) = (1/12) / (1/2)$
 $P(B|A) = \frac{1}{6}$
- $P(B|A) = P(B) = \frac{1}{6}$ as they are conditionally independent.

Naive Bayes

- $P(y|x_1, x_2) = \frac{P(x_1|y) \times P(x_2|y) \times P(y)}{P(x_1) \times P(x_2)}$
- $x_1 = \text{Good}; x_2 = \text{Bad}$
- $P(Y | \text{Good, Bad}) = \frac{P(\text{Good}|Y) \times P(\text{Bad}|Y) \times P(Y)}{P(\text{Good}) \times P(\text{Bad})}$

x1	x2	y
Mark s	Extra Curriculars	Admission Status
Good	Bad	Y
Bad	Bad	N
Bad	Good	Y

Naive Bayes

x1	Yes	No	$P(x1 Y)$	$P(x1 N)$
Good	1	0	1/2	0/1
Bad	1	1	1/2	1/1
Total	2	1	1	1

x2	Yes	No	$P(x2 Y)$	$P(x2 N)$
Good				
Bad				
Total			1	1

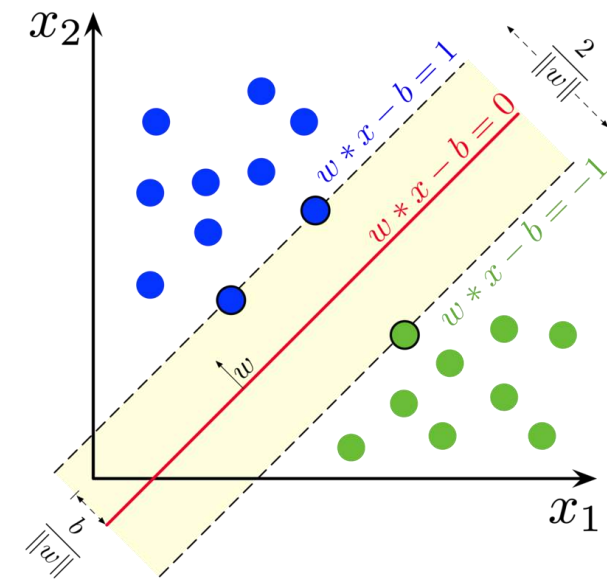
Naive Bayes

x1	x2	y
Mark s	Extra Curriculars	Admission Status
Good	Bad	Y
Bad	Bad	N
Bad	Good	Y

- $P(Y \mid \text{Good, Bad}) = \frac{P(\text{Good} \mid Y) \times P(\text{Bad} \mid Y) \times P(Y)}{P(\text{Good}) \times P(\text{Bad})} = \frac{\frac{1}{2} \times \frac{1}{2} \times \frac{2}{3}}{\frac{1}{3} \times \frac{2}{3}}$
- = 3/4 or 75% probability.

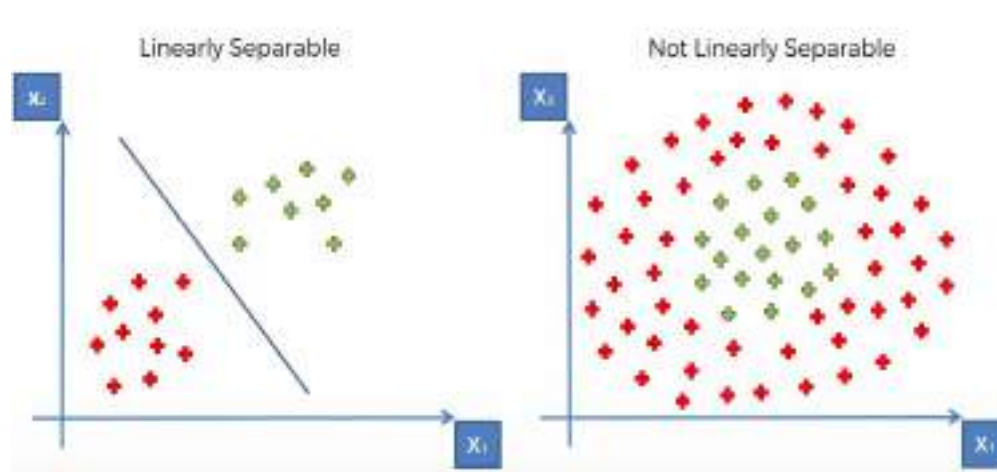
SVM

- SVM stands for Support Vector Machines, Where the main agenda of the models is to split the data into n categories, n being the number of categories present. The basic version of the SVM uses a linear hyperplane to distinguish between 2 or more classes.
- A hyperplane is nothing but a line (2D) or a plane (3D) that is used to determine which side of it will it fall into when a new point is provided to it.



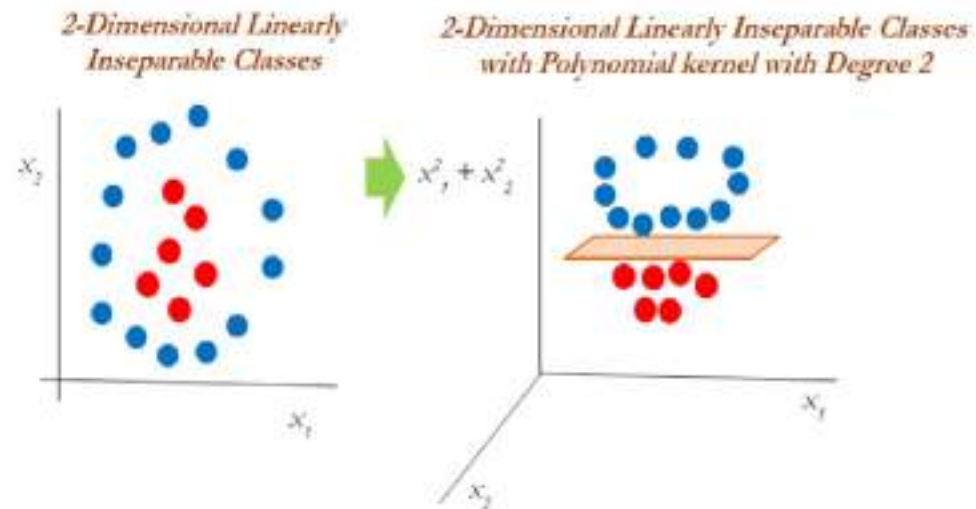
What are the steps to create a SVM model?

- The Process is simple:
- Step1 : Identify the support vectors.
- Step2 : See if the data is linearly separable.
- Step3 : If it is then make linear separator between them (exactly in the middle).
- Step4 : your model is ready.



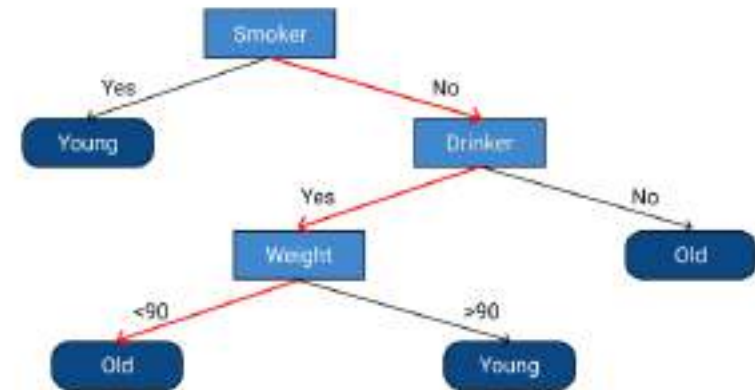
What are the steps for non linearly separable data points?

- The Process is simple:
- Step1: Use the kernel function to take the data points to a higher dimension.
- Step2 : Identify the support vectors.
- Step3 : Draw the hyperplane in N dimensions.
- Step4 : your model is ready.



CART

- CART stands for Classification and Regression Trees. The classification trees are also referred as decision trees. The name of the model is decision tree because it looks like an inverted tree. The decision is made on the basis of some condition and the splits are made accordingly.
- The tree goes on till the last node (referred as the leaf node), till it can't be split any further.

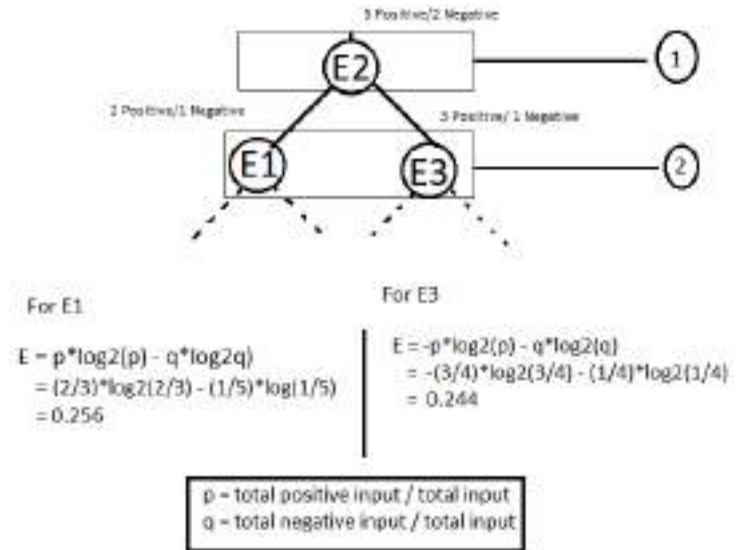


What is the criteria for splitting?

- The root node(starting point) is calculated using the information gain. The information gain is calculated in 2 ways, entropy and gini.

- Entropy is calculated by:

$$E = -p*\log_2(p) - q*\log_2(q)$$



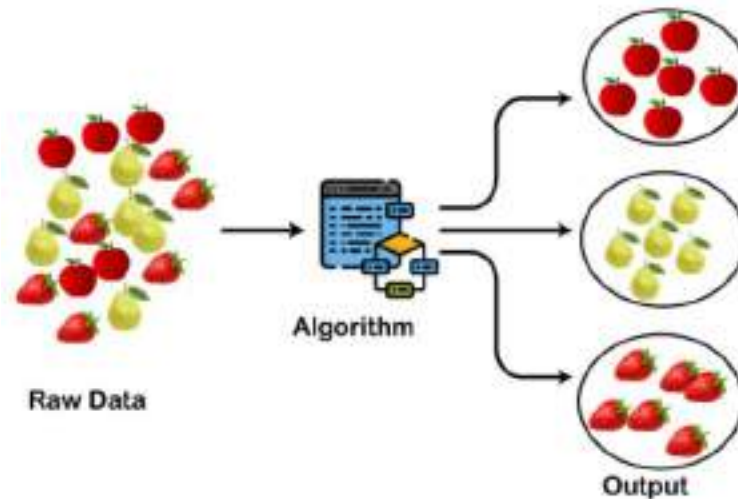
Disadvantages of Decision Trees:

Decision trees to the max depth tend to overfit. (Generally)

- **Overfit:** when the model takes in the training values and fit it perfectly and fails to perform well on the testing set, we say that our model has overfit. You can test by checking the training and testing accuracy. If the training accuracy is way higher than the testing accuracy, it is safe to say that the model did over fit.
- **Underfit:** when the data is unable to fit well, ie. the accuracy is lower on both the training and testing set we go on to say that the model underfit. This is generally the case when the model we are using is too simple for the data set.

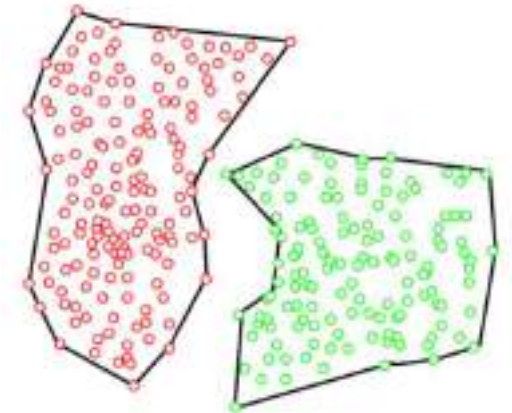
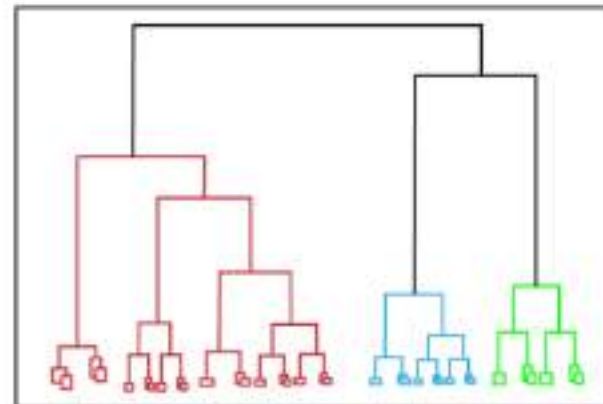
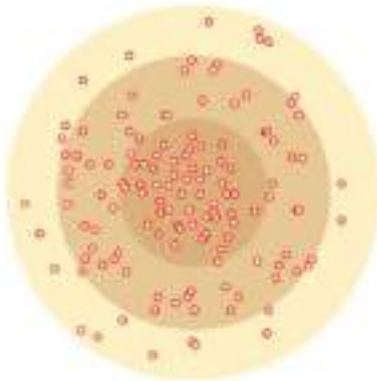
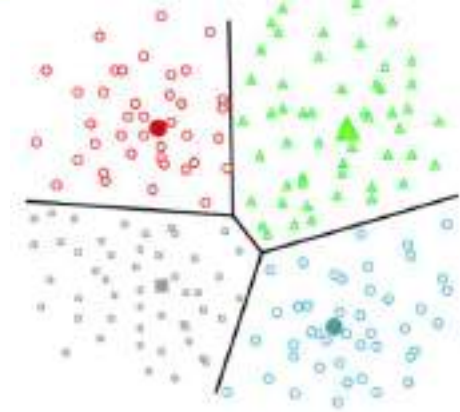
Clustering

- A way of grouping the data points into different clusters, consisting of similar data points. The objects with the possible similarities remain in a group that has less or no similarities with another group.
- The clustering technique is commonly used for statistical data analysis.



Types of Clustering Methods

- Partitioning Clustering
- Density-Based Clustering
- Distribution Model-Based Clustering
- Hierarchical Clustering
- Fuzzy Clustering



Applications of Clustering



- In Identification of Cancer Cells
- Search Engines
- Customer Segmentation
- Biology
- Land Use

K means Clustering

- We choose an arbitrary k value which depicts the number of centroids, which in turn depicts the number of clusters.
- The distance from centroids to all the data points is calculated and the centroid with the least distance is assigned that that point.
- The cycle goes on till the the centroid values don't change significantly.

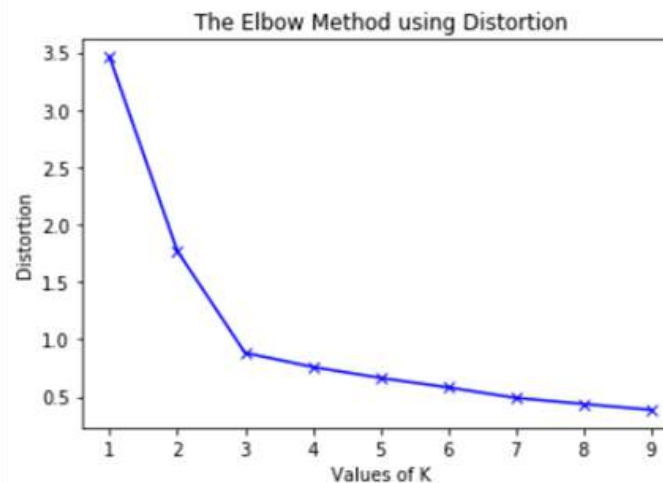


How does the K-Means Algorithm Work?

- Step-1: Select the number K to decide the number of clusters.
- Step-2: Select random K points or centroids. (It can be other from the input dataset).
- Step-3: Assign each data point to their closest centroid, which will form the predefined K clusters.
- Step-4: Calculate the variance and place a new centroid of each cluster.
- Step-5: Repeat the third steps, which means reassign each datapoint to the new closest centroid of each cluster.
- Step-6: If any reassignment occurs, then go to step-4 else go to FINISH.
- Step-7: The model is ready.

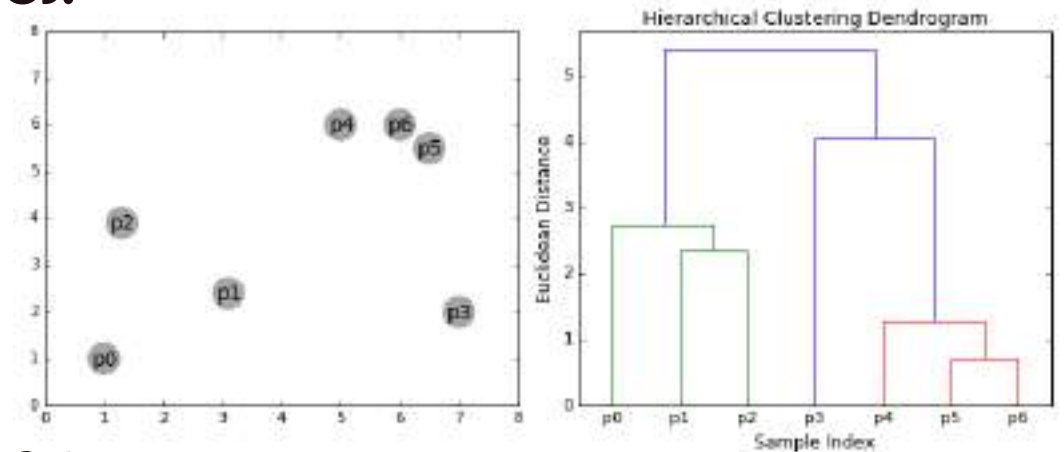
How to find the optimal value for K in K means?

- We use something called the elbow graph. The graph plots the value of k against the amount of distortion in the data points. The point after which the graph plateaus, is called the elbow point.
- In the example we can see the elbow form at $k=3$.



Hierarchical Clustering

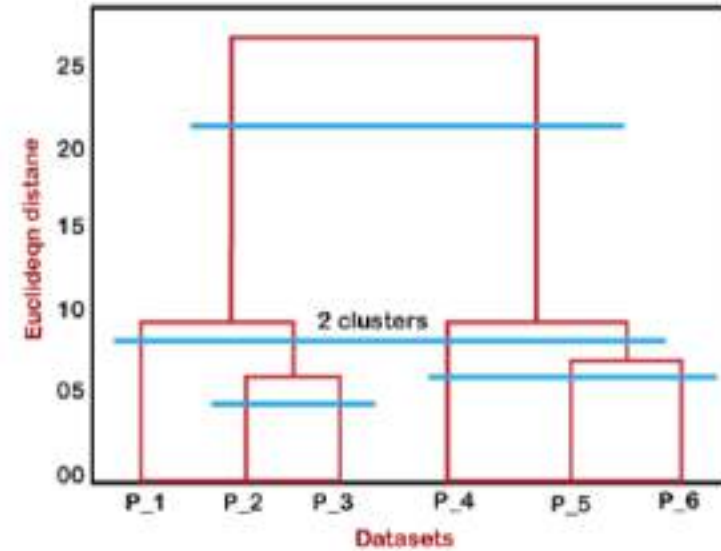
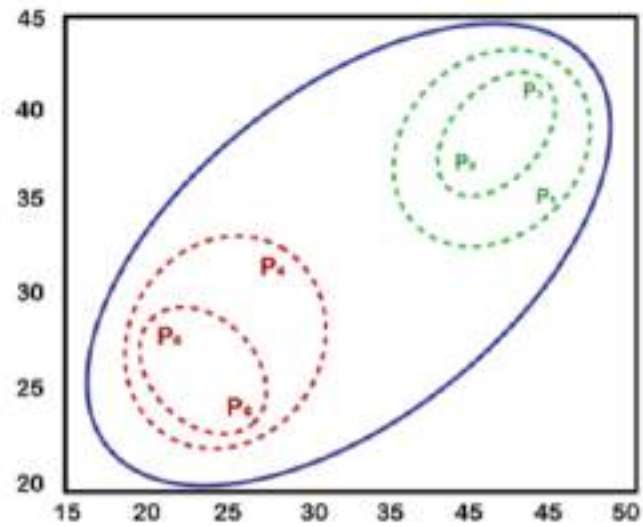
- We use something called a dendrogram for this purpose. There are 2 approaches:
- Top down approach: Consider all the data points to be in one single cluster, and then go on splitting.
- Bottom up approach: Consider all the data points to be single clusters and then club them as you move up.
- Now in each case the distance is calculated and the ones with the least distance are clubbed together.



How do linkage affect the dendrogram ?

- There are 3 kind of linkage that can be used:
- **Single Linkage:** For two clusters $C1$ and $C2$, the single linkage returns the minimum distance between two points i and j such that i belongs to $C1$ and j belongs to $C2$.
- **Complete Linkage:** For two clusters $C1$ and $C2$, the complete linkage returns the maximum distance between two points i and j such that i belongs to $C1$ and j belongs to $C2$.
- **Average Linkage:** For two clusters $C1$ and $C2$, first for the distance between any data-point i in $C1$ and any data-point j in $C2$ and then the arithmetic mean of these distances are calculated.

Working of Dendrogram in Hierarchical clustering



DBSCAN

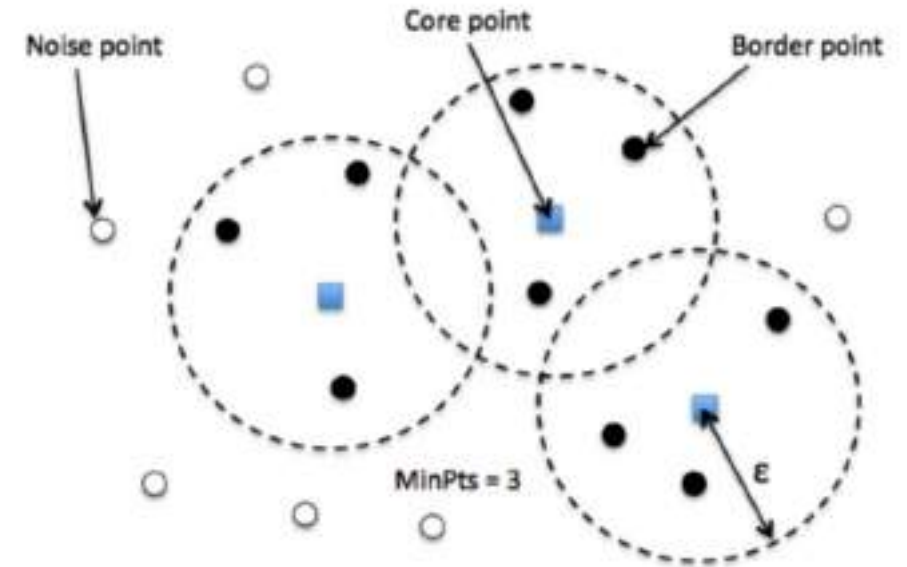
- **Density-Based Clustering** identifies distinctive groups/clusters in the data, based on the idea that a cluster in data space is a contiguous region of high point density, separated from other such clusters by contiguous regions of low point density.
- **DBSCAN** can discover clusters of different shapes and sizes from a large amount of data, which is containing noise and outliers.

DBSCAN

- The DBSCAN algorithm uses two parameters:
- minPts: The minimum number of points (a threshold) clustered together for a region to be considered dense.
- eps (ϵ): A distance measure that will be used to locate the points in the neighborhood of any point.
- Reachability in terms of density establishes a point to be reachable from another if it lies within a particular distance (eps) from it.
- Connectivity, on the other hand, involves a transitivity based chaining-approach to determine whether points are located in a particular cluster.

DBSCAN

- Core — This is a point that has at least m points within distance n from itself.
- Border — This is a point that has at least one Core point at a distance n .
- Noise — This is a point that is neither a Core nor a Border. And it has less than m points within distance n from itself.

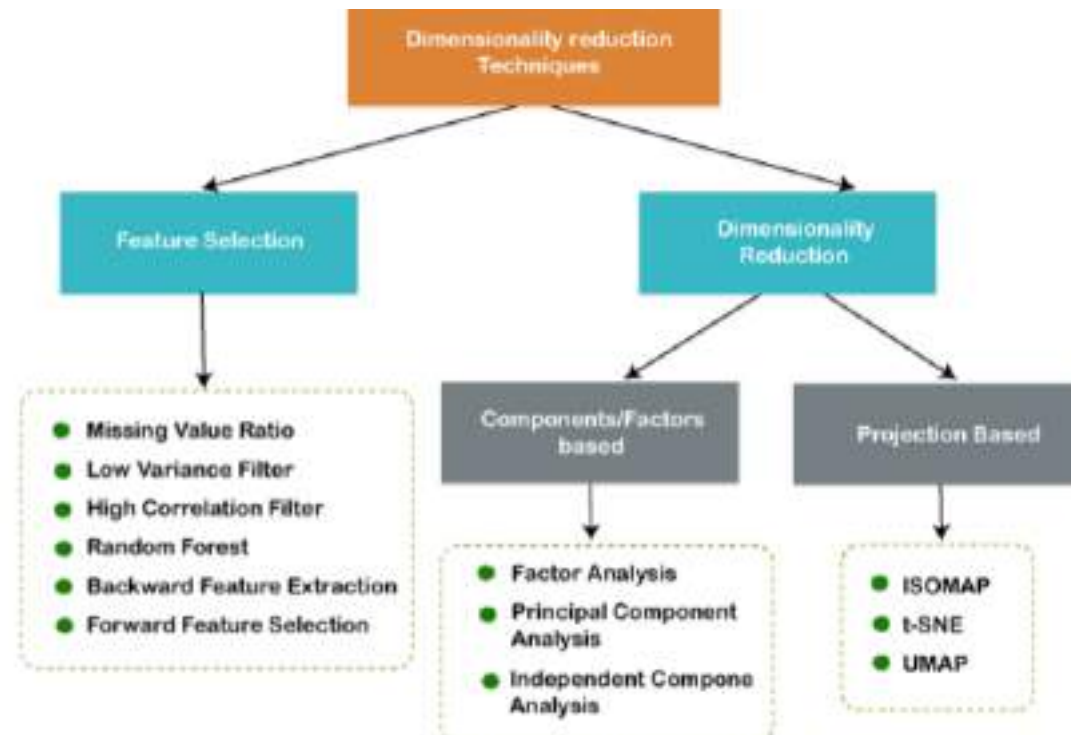


Algorithmic steps for DBSCAN clustering

- The algorithm proceeds by arbitrarily picking up a point in the dataset (until all points have been visited).
- If there are at least 'minPoint' points within a radius of ' ϵ ' to the point then we consider all these points to be part of the same cluster.
- The clusters are then expanded by recursively repeating the neighborhood calculation for each neighboring point

Dimensionality Reduction

- It is a way of converting the higher dimensions dataset into lesser dimensions dataset ensuring that it provides similar information.
- It is commonly used in the fields that deal with high-dimensional data.



Dimensionality Reduction

Benefits

- By reducing the dimensions of the features, the space required to store the dataset also gets reduced.
- Less Computation training time is required for reduced dimensions of features.
- Reduced dimensions of features of the dataset help in visualizing the data quickly.
- It removes the redundant features (if present) by taking care of multicollinearity.

Dimensionality Reduction



Disadvantages

- Some data may be lost due to dimensionality reduction.
- In the PCA dimensionality reduction technique, sometimes the principal components required to consider are unknown.

Dimensionality Reduction



Feature Selection

- Filters Methods
- Wrappers Methods
- Embedded Methods

Feature Extraction

- Principal Component Analysis
- Linear Discriminant Analysis
- Kernel PCA
- Quadratic Discriminant Analysis

Common techniques of Dimensionality Reduction

- **Principal Component Analysis**
- **Backward Elimination**
- **Forward Selection**
- **Missing Value Ratio**
- **Low Variance Filter**
- **High Correlation Filter**
- **Random Forest**
- **Factor Analysis**
- **Auto-Encoder**

Principal Component Analysis

- It is a statistical process that converts the observations of correlated features into a set of linearly uncorrelated features with the help of orthogonal transformation.
- These new transformed features are called the Principal Components.
- PCA generally tries to find the lower-dimensional surface to project the high-dimensional data.
- The PCA algorithm is based on some mathematical concepts such as:
 - Variance and Covariance
 - Eigenvalues and Eigen factors

Some common terms used in PCA algorithm

- Dimensionality
- Correlation
- Orthogonal
- Eigenvectors
- Covariance Matrix

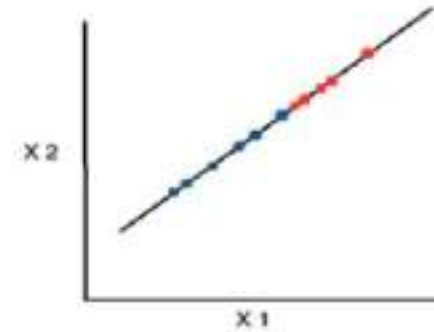
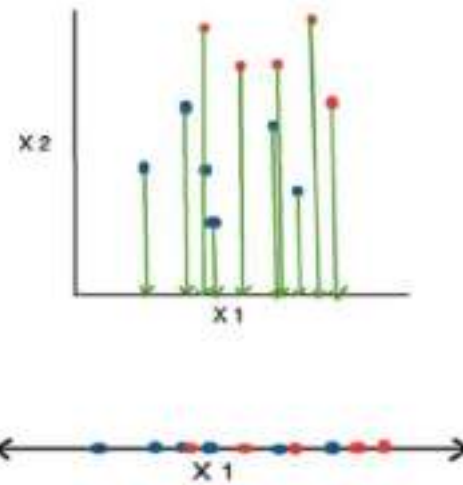
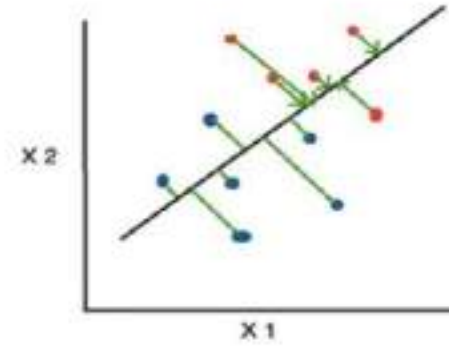
Steps for PCA algorithm

- Getting the dataset
- Representing data into a structure
- Standardizing the data
- Calculating the Covariance of Z
- Calculating the Eigen Values and Eigen Vectors
- Sorting the Eigen Vectors
- Calculating the new features Or Principal Components
- Remove less or unimportant features from the new dataset.

Linear Discriminant Analysis

- It is used as a pre-processing step in Machine Learning and applications of pattern classification.
- The goal of LDA is to project the features in higher dimensional space onto a lower-dimensional space in order to avoid the curse of dimensionality and also reduce resources and dimensional costs.
- Limitations of Logistic Regression
 - Two-class problems
 - Unstable with Well-Separated classes
 - Unstable with few examples

Linear Discriminant Analysis



How does LDA work?

- We need to calculate the separability between classes which is the distance between the mean of different classes. This is called the between-class variance.
- Calculate the distance between the mean and sample of each class. It is also called the within-class variance.
- Construct the lower-dimensional space which maximizes the between-class variance and minimizes the within-class variance. P is considered as the lower-dimensional space projection, also called Fisher's criterion.

How to prepare data from LDA?

- LDA is mainly used in classification problems where you have a categorical output variable. It allows both binary classification and multi-class classification.
- The standard LDA model makes use of the Gaussian Distribution of the input variables. You should check the univariate distributions of each attribute and transform them into a more Gaussian-looking distribution. Outliers can skew the primitive statistics used to separate classes in LDA, so it is preferable to remove them.
- Since LDA assumes that each input variable has the same variance, it is always better to standardize your data before using an LDA model. Keep the mean to be 0 and the standard deviation to be 1.

Recommendation Systems



- Recommendation engines are a subclass of machine learning which generally deal with ranking or rating products / users.
- They're used by various large name companies like Google, Instagram, Spotify, Amazon, Reddit, Netflix etc. often to increase engagement with users and the platform.
- Recommender systems are often seen as a “black box”, the model created by these large companies are not very easily interpretable.

What Defines a Good Recommendation?

- The quality of a recommendation can be assessed through various tactics which measure coverage and accuracy.
- Accuracy is the fraction of correct recommendations out of total possible recommendations while coverage measures the fraction of objects in the search space the system is able to provide recommendations for.
- Recommender systems share several conceptual similarities with the classification and regression modelling problem.
- K Fold Cross Validation
- MAE (Mean Absolute Error)
- RMSE (Root Mean Square Deviation)

Collaborative Filtering Systems

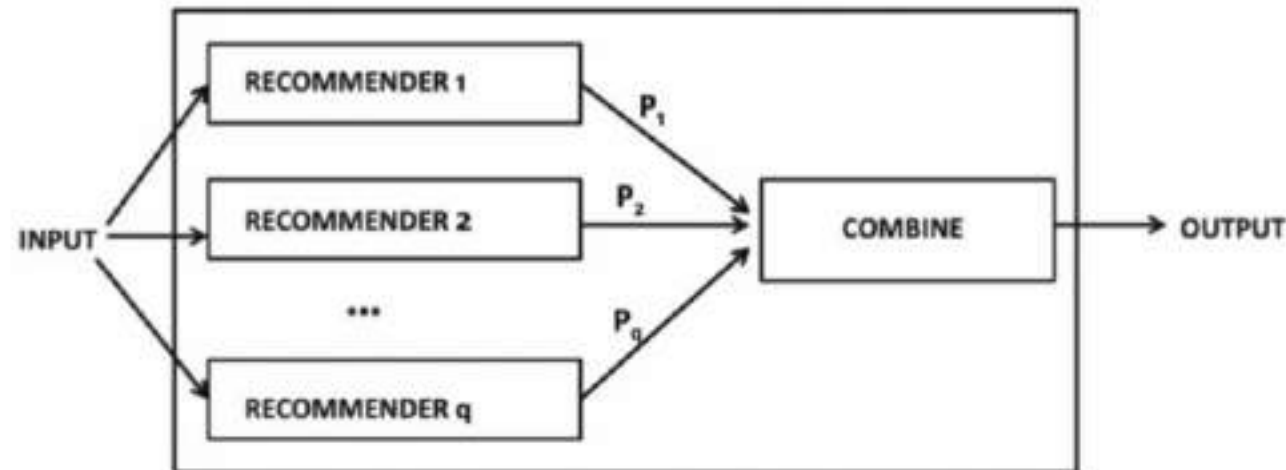
- Collaborative filtering is the process of predicting the interests of a user by identifying preferences and information from many users.
- There are two common types of approaches in collaborative filtering, memory based and model based approach.
- Examples:
 - YouTube content recommendation to users
 - Coursera course recommendation

Content Based Systems

- Content based systems generate recommendations based on the users preferences and profile.
- Unlike most collaborative filtering models which leverage ratings between target user and other users, content based models focus on the ratings provided by the target user themselves.
- The simplest forms of content based systems require the following sources of data
 - Item level data source
 - User level data source
- Examples
 - Amazon product feed
 - Spotify music recommendations

Hybrid Recommendation System

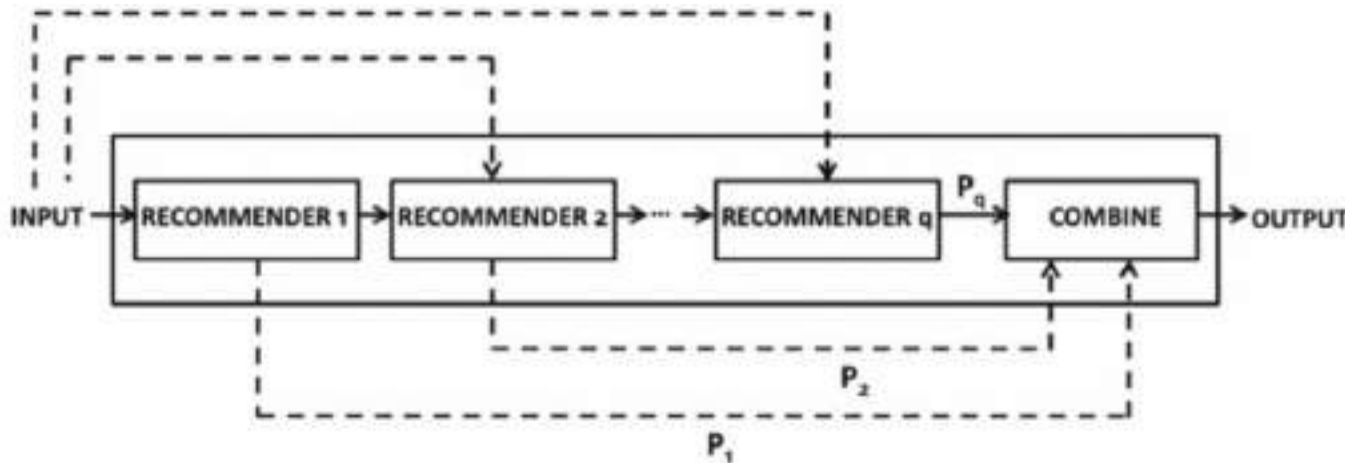
- Hybrid recommender systems are ones designed to use different available data sources to generate robust inferences.
- The parallel design provides the input to multiple recommendation systems, each of those recommendations are combined to generate one output.



(a) Parallel design

Hybrid Recommendation System

- The sequential design provides the input parameters to a single recommendation engine, the output is passed on to the following recommender in a sequence.



(b) Sequential design

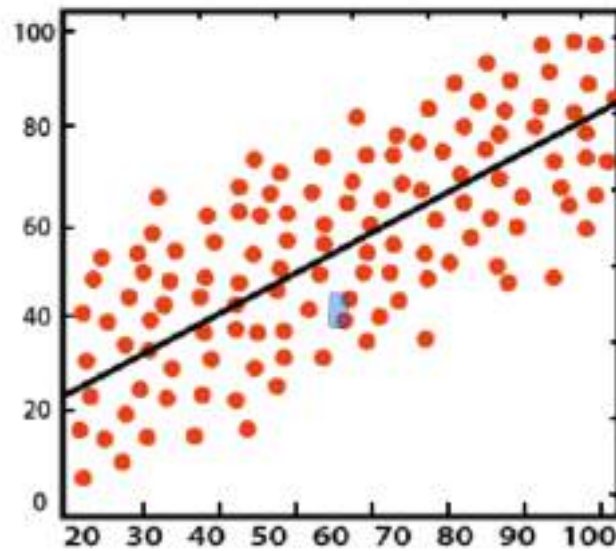
Difference between Supervised and Unsupervised Learning

Supervised Learning	Unsupervised Learning
Supervised learning algorithms are trained using labeled data.	Unsupervised learning algorithms are trained using unlabeled data.
Supervised learning model takes direct feedback to check if it is predicting correct output or not.	Unsupervised learning model does not take any feedback.
Supervised learning model predicts the output.	Unsupervised learning model finds the hidden patterns in data.
In supervised learning, input data is provided to the model along with the output.	In unsupervised learning, only input data is provided to the model.
The goal of supervised learning is to train the model so that it can predict the output when it is given new data.	The goal of unsupervised learning is to find the hidden patterns and useful insights from the unknown dataset.
Supervised learning needs supervision to train the model.	Unsupervised learning does not need any supervision to train the model.
Supervised learning can be categorized in Classification and Regression problems.	Unsupervised Learning can be classified in Clustering and Associations problems.

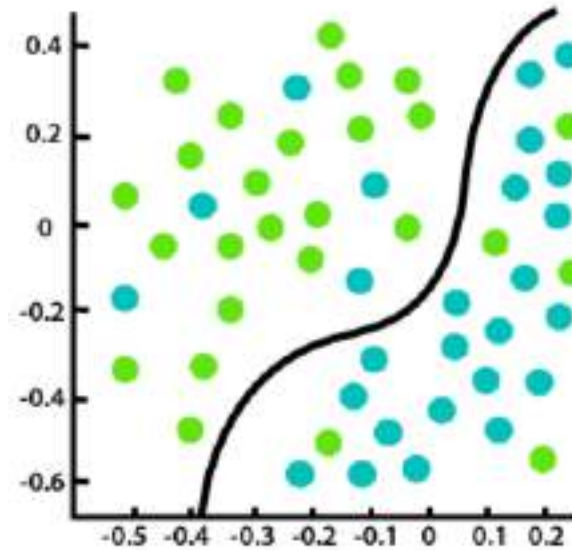
Difference between Supervised and Unsupervised Learning

Supervised Learning	Unsupervised Learning
Supervised learning can be used for those cases where we know the input as well as corresponding outputs.	Unsupervised learning can be used for those cases where we have only input data and no corresponding output data.
Supervised learning model produces an accurate result.	Unsupervised learning model may give less accurate result as compared to supervised learning.
Supervised learning is not close to true Artificial intelligence as in this, we first train the model for each data, and then only it can predict the correct output.	Unsupervised learning is more close to the true Artificial Intelligence as it learns similarly as a child learns daily routine things by his experiences.
It includes various algorithms such as Linear Regression, Logistic Regression, Support Vector Machine, Multi-class Classification, Decision tree, Bayesian Logic, etc.	It includes various algorithms such as Clustering, KNN, and Apriori algorithm.

Regression vs. Classification



Regression



Classification

Regression vs. Classification

Regression Algorithm	Classification Algorithm
In Regression, the output variable must be of continuous nature or real value.	In Classification, the output variable must be a discrete value.
The task of the regression algorithm is to map the input value (x) with the continuous output variable(y).	The task of the classification algorithm is to map the input value(x) with the discrete output variable(y).
Regression Algorithms are used with continuous data.	Classification Algorithms are used with discrete data.
In Regression, we try to find the best fit line, which can predict the output more accurately.	In Classification, we try to find the decision boundary, which can divide the dataset into different classes.
Regression algorithms can be used to solve the regression problems such as Weather Prediction, House price prediction, etc.	Classification Algorithms can be used to solve classification problems such as Identification of spam emails, Speech Recognition, Identification of cancer cells, etc.
The regression Algorithm can be further divided into Linear and Non-linear Regression.	The Classification algorithms can be divided into Binary Classifier and Multi-class Classifier.

Semi-Supervised Learning



- Semi-Supervised represents the intermediate ground between Supervised and Unsupervised learning algorithms. It uses the combination of labeled and unlabeled datasets during the training period.
- To overcome these drawbacks of supervised learning and unsupervised learning algorithms, the concept of Semi-supervised learning is introduced.
- Under semi-supervised learning, the student has to revise itself after analyzing the same concept under the guidance of an instructor at college.

Assumptions followed by Semi-Supervised Learning

- **Continuity Assumption:** As per the continuity assumption, the objects near each other tend to share the same group or label.
- **Cluster assumptions-** In this assumption, data are divided into different discrete clusters. Further, the points in the same cluster share the output label.
- **Manifold assumptions-** This assumption helps to use distances and densities, and this data lie on a manifold of fewer dimensions than input space. The dimensional data are created by a process that has less degree of freedom and may be hard to model directly. (This assumption becomes practical if high).

Working of Semi-Supervised Learning

- It trains the model with less amount of training data similar to the supervised learning models. The training continues until the model gives accurate results.
- The algorithms use the unlabeled dataset with pseudo labels in the next step, and now the result may not be accurate.
- Now, the labels from labeled training data and pseudo labels data are linked together.
- The input data in labeled training data and unlabeled training data are also linked.
- In the end, again train the model with the new combined input as did in the first step. It will reduce errors and improve the accuracy of the model.

Real-world applications of Semi-supervised Learning

- **Speech Analysis**
- **Web content classification**
- **Protein sequence classification**
- **Text document classifier**

Reinforcement Learning

- Reinforcement Learning is a feedback-based Machine learning technique in which an agent learns to behave in an environment by performing the actions and seeing the results of actions. For each good action, the agent gets positive feedback, and for each bad action, the agent gets negative feedback or penalty.
- In Reinforcement Learning, the agent learns automatically using feedbacks without any labeled data, unlike supervised learning.
- Since there is no labeled data, so the agent is bound to learn by its experience only.

Terms used in Reinforcement Learning

- Agent
- Environment
- Action
- State
- Reward
- Policy
- Value
- Q-value

Key Features of Reinforcement Learning

- In RL, the agent is not instructed about the environment and what actions need to be taken.
- It is based on the hit and trial process.
- The agent takes the next action and changes states according to the feedback of the previous action.
- The agent may get a delayed reward.
- The environment is stochastic, and the agent needs to explore it to reach to get the maximum positive rewards.

Approaches to implement Reinforcement Learning



- **Value-based:** The value-based approach is about to find the optimal value function, which is the maximum value at a state under any policy. Therefore, the agent expects the long-term return at any state(s) under policy π .
- **Policy-based:** Policy-based approach is to find the optimal policy for the maximum future rewards without using the value function. In this approach, the agent tries to apply such a policy that the action performed in each step helps to maximize the future reward

Approaches to implement Reinforcement Learning



- **Policy-based:** The policy-based approach has mainly two types of policy:
 - **Deterministic:** The same action is produced by the policy (π) at any state.
 - **Stochastic:** In this policy, probability determines the produced action.
- **Model-based:** In the model-based approach, a virtual model is created for the environment, and the agent explores that environment to learn it. There is no particular solution or algorithm for this approach because the model representation is different for each environment.

Types of Reinforcement learning

- **Positive Reinforcement:**

- The positive reinforcement learning means adding something to increase the tendency that expected behavior would occur again. It impacts positively on the behavior of the agent and increases the strength of the behavior. This type of reinforcement can sustain the changes for a long time, but too much positive reinforcement may lead to an overload of states that can reduce the consequences.

- **Negative Reinforcement:**

- The negative reinforcement learning is opposite to the positive reinforcement as it increases the tendency that the specific behavior will occur again by avoiding the negative condition. It can be more effective than the positive reinforcement depending on situation and behavior, but it provides reinforcement only to meet minimum behavior.

Reinforcement Learning Algorithms

- **Q-Learning:**
 - Q-learning is an Off policy RL algorithm, which is used for the temporal difference Learning. The temporal difference learning methods are the way of comparing temporally successive predictions.
- **State Action Reward State action (SARSA):**
 - SARSA stands for State Action Reward State action, which is an on-policy temporal difference learning method. The on-policy control method selects the action for each state while learning using a specific policy.
- **Deep Q Neural Network (DQN):**
 - As the name suggests, DQN is a Q-learning using Neural networks. For a big state space environment, it will be a challenging and complex task to define and update a Q-table.

Reinforcement Learning Applications



- **Robotics**
- **Control**
- **Game Playing**
- **Chemistry**
- **Business**
- **Manufacturing**
- **Finance Sector**